NxM Templates for SuperCDMS SNOLAB Run 4 — Ge Activation Data

K-line event selection · free-pretrigger two-exponential fits *(future step: three-, four-exponential fits)* · 1x1 and NxM (PCA) templates

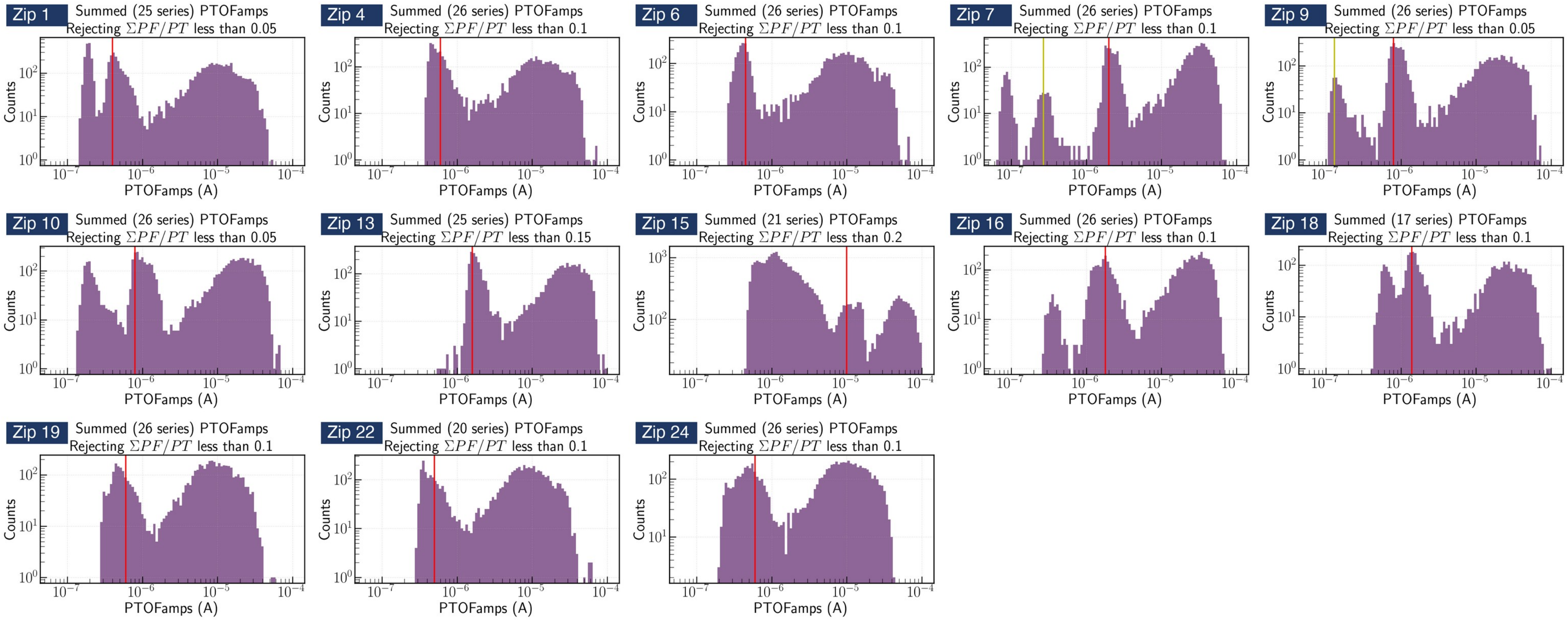
Zhiheng Li — July 2026

Table of Contents

Ge activation dataset - R4

Credit: Prof. Tarek Saab

- Per detector, Prof. Saab's study marked the 10.37 keV K-line position in **PTOFamps** (the red line in each panel below)
- A **PTOFamps** window bracketing the red line: position ×/÷ 1.35, a rough eyeballed choice; deliberately minimal, no other cuts
- For every selected event: cache the fully unprocessed raw MIDAS traces of all channels + event metadata, saved as per-series pickle files (reusable for any later analysis)
- Events in the window per zip: Z1 10.1k Z4 26.1k Z6 17.0k Z7 2.2k Z9 2.6k Z10 7.6k Z13 9.1k Z15 4.2k Z16 1.6k Z18 22.1k Z19 23.3k Z22 25.0k Z24 22.9k (total 174k)
- **This talk focuses on Z7, the best detector; all other detectors are covered in the backup slides**



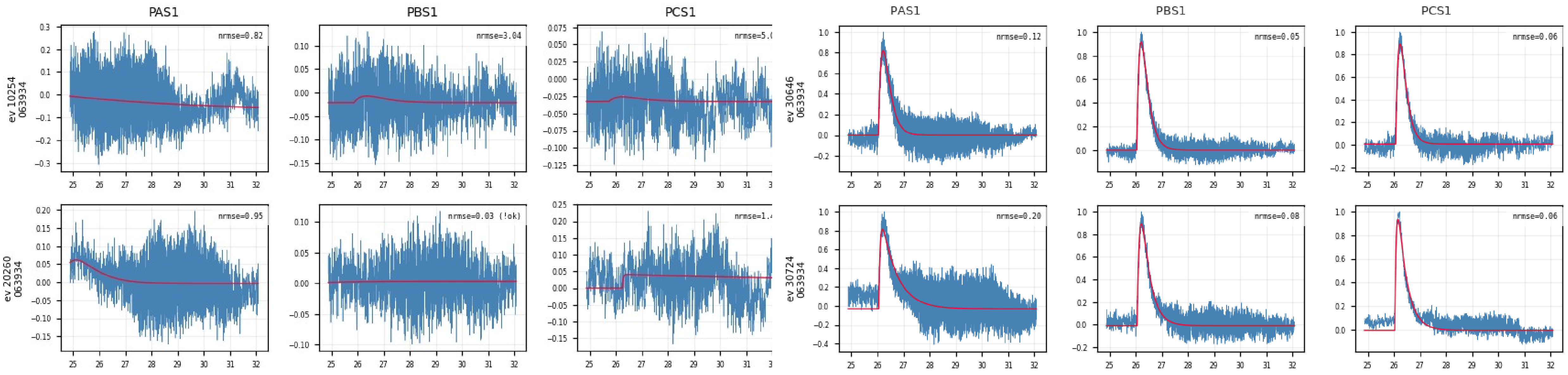
All 13 detectors, PTOFamps spectra (Prof. Saab's study)

Fit quality at a glance: event × channel grids

• Real events and noise triggers can be told apart by visual comparison (no cut applied at this stage)

NOISE

REAL PULSE



Per-trace algorithm: low-pass → fit → align

1. **Low-pass** — 100 kHz Butterworth
2. **Normalize** — subtract the baseline (median of the pre-pulse samples), scale the trace to unit peak
3. **Fit** — two-exponential model:

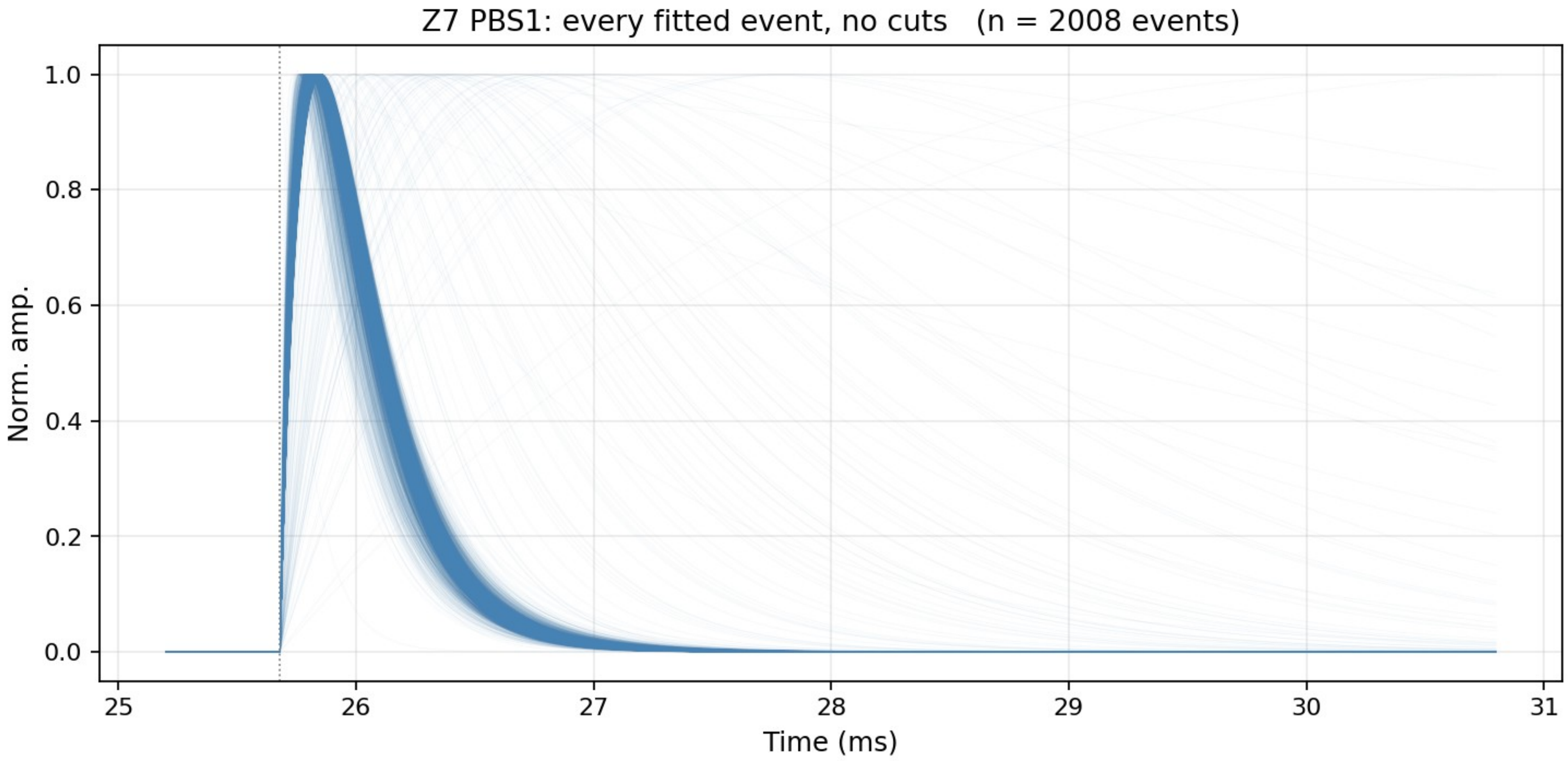
$$y(t) = A[\,e^{-(t-t_0)/\tau_{\mathrm{fall}}} - e^{-(t-t_0)/\tau_{\mathrm{rise}}}\,] + b$$

$t_0 =$ pretrigger, free within 16050 ± 3000 samples (τ_{rise}, τ_{fall}: rise / fall time)

Fit parameters: A amplitude · τ_rise rise time constant · τ_fall fall time constant · t_0 pulse onset (pretrigger) · b baseline

4. **Two quality checks per trace** (calculated and recorded here, not cut yet):
   - **fit_ok** — amplitude is positive, and the pulse rises faster than it falls
   - **NRMSE** — RMS of the fit residual ÷ pulse peak (smaller = better fit)

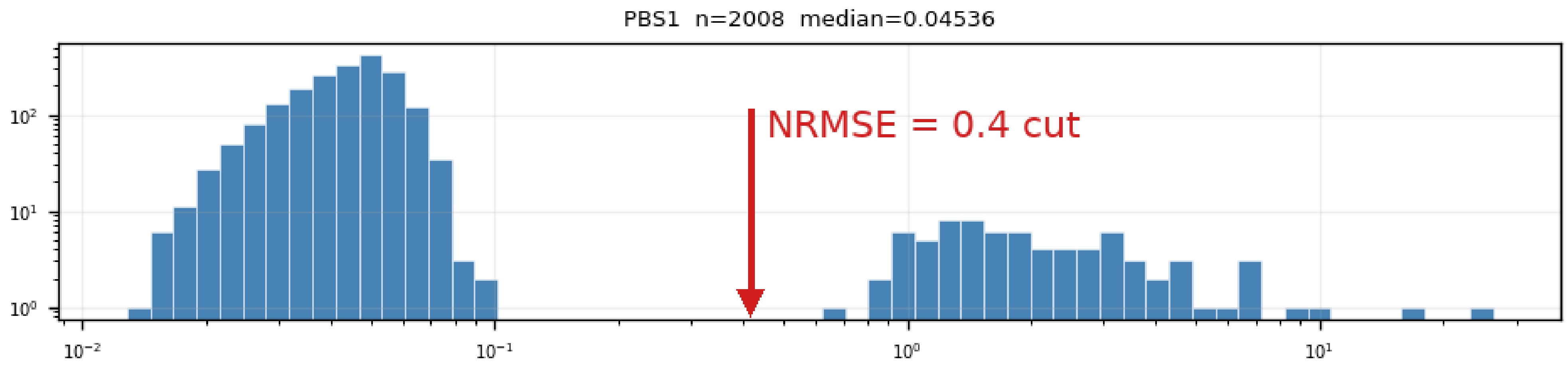
5. Align



Z7 PBS1: every fitted event drawn at the common pretrigger, no cuts of any kind (n = 2008 events)

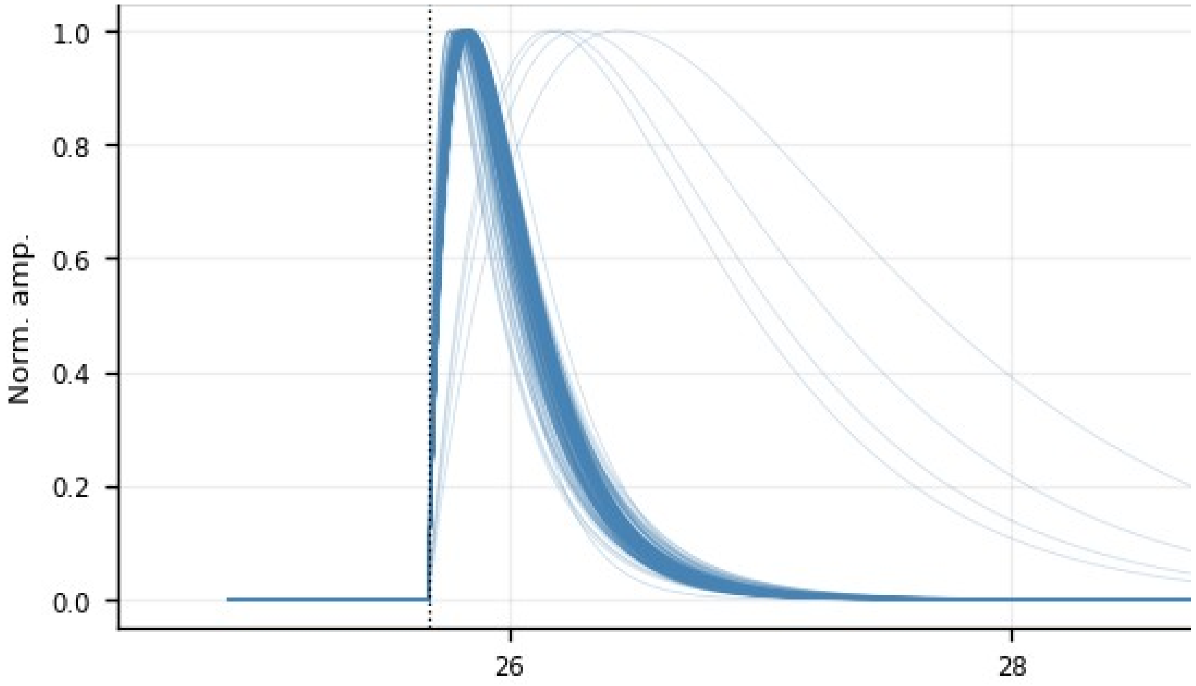
Quality cut 1: NRMSE ≤ 0.4 — where the number comes from

- NRMSE of fit_ok events is bimodal: good fits (median ≈ 0.05–0.1) vs noise triggers (≈ 1–2), valley at ≈ 0.4–0.5
- The cut sits in the valley — it is read off the distribution, not tuned on the templates
- Weak detectors (Z1, Z4, Z6, Z18, Z19, Z22, Z24): same bimodal picture, noise peak dominates

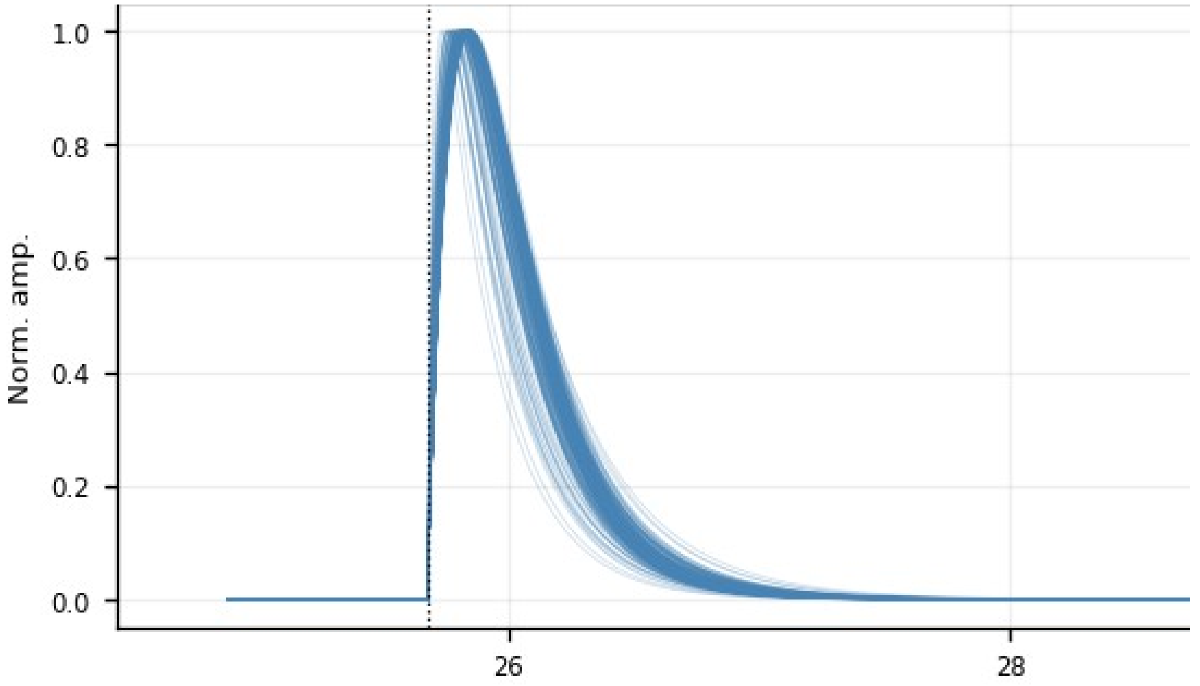


Z7 PBS1: two clean populations, log-log axes

Aligned fitted curves: before and after the NRMSE cut



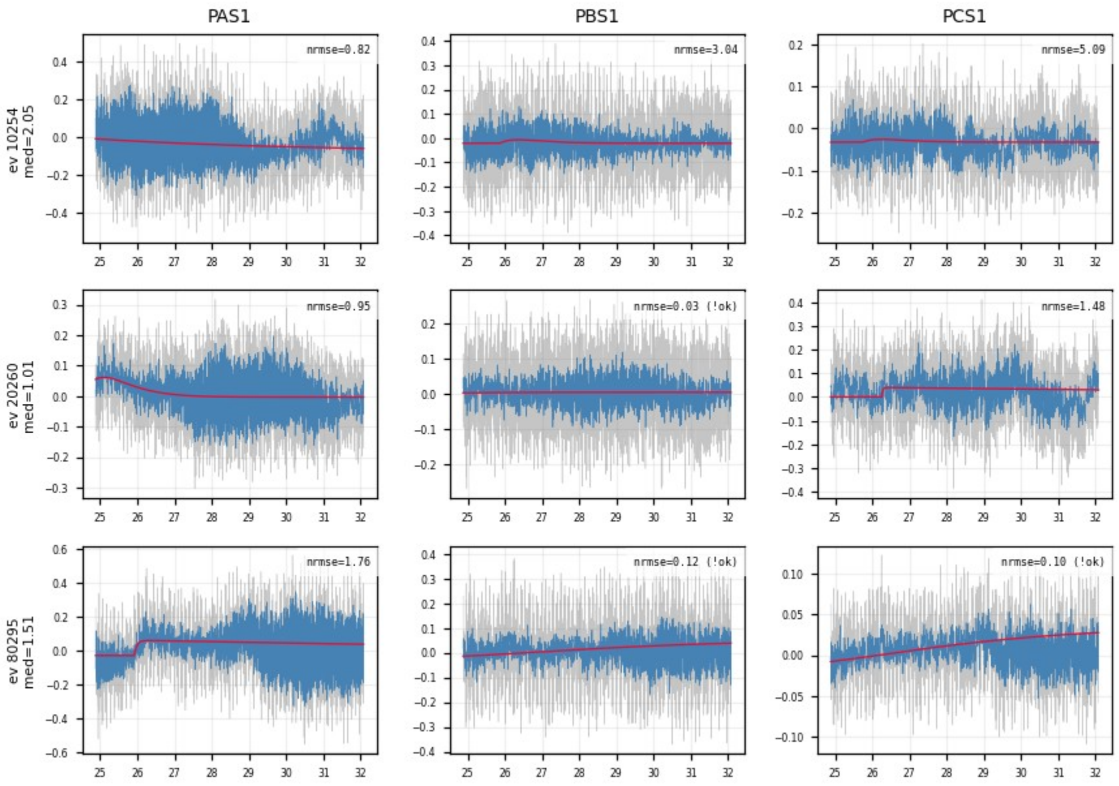
All fit_ok fitted curves (Z7 PBS1), 200-curve sample drawn



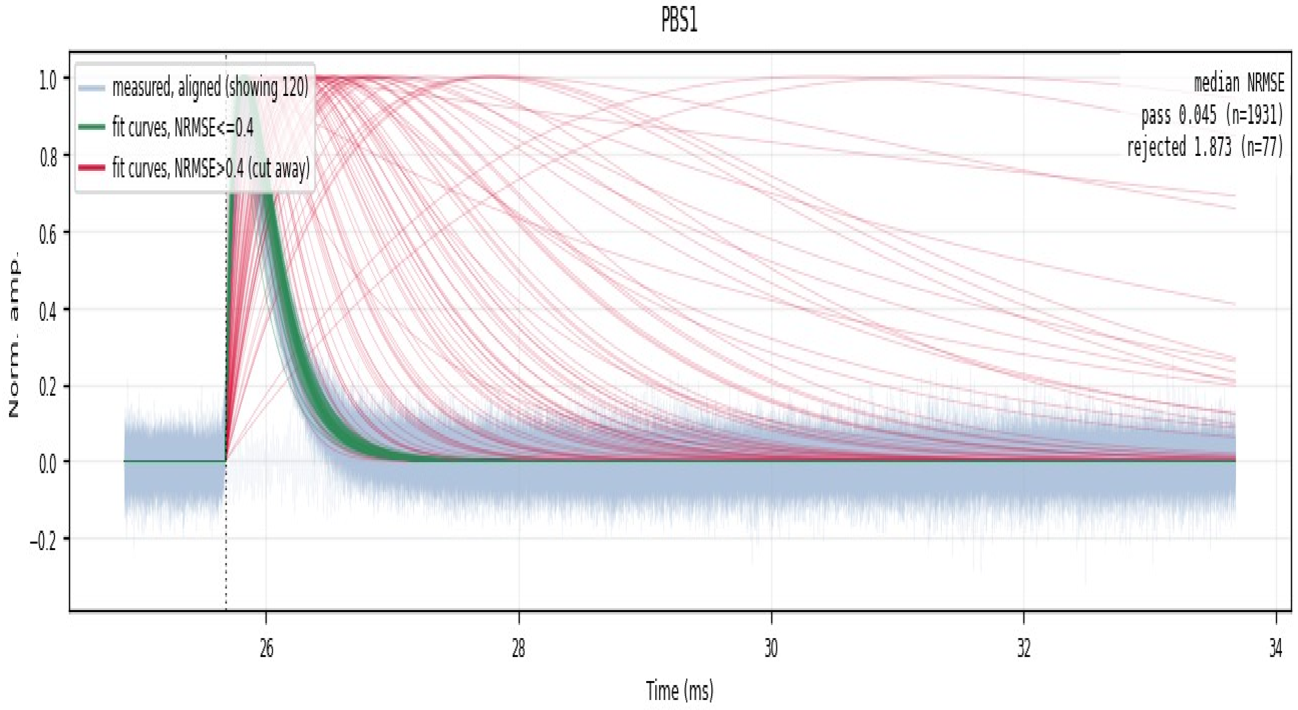
After NRMSE ≤ 0.4 (Z7 PBS1): one tight shape family

The rejected population is noise — verified in the raw traces

• Event grids of NRMSE-rejected events (median NRMSE > 0.4 across channels): the raw traces show no pulse — the cut removes noise triggers, not physics



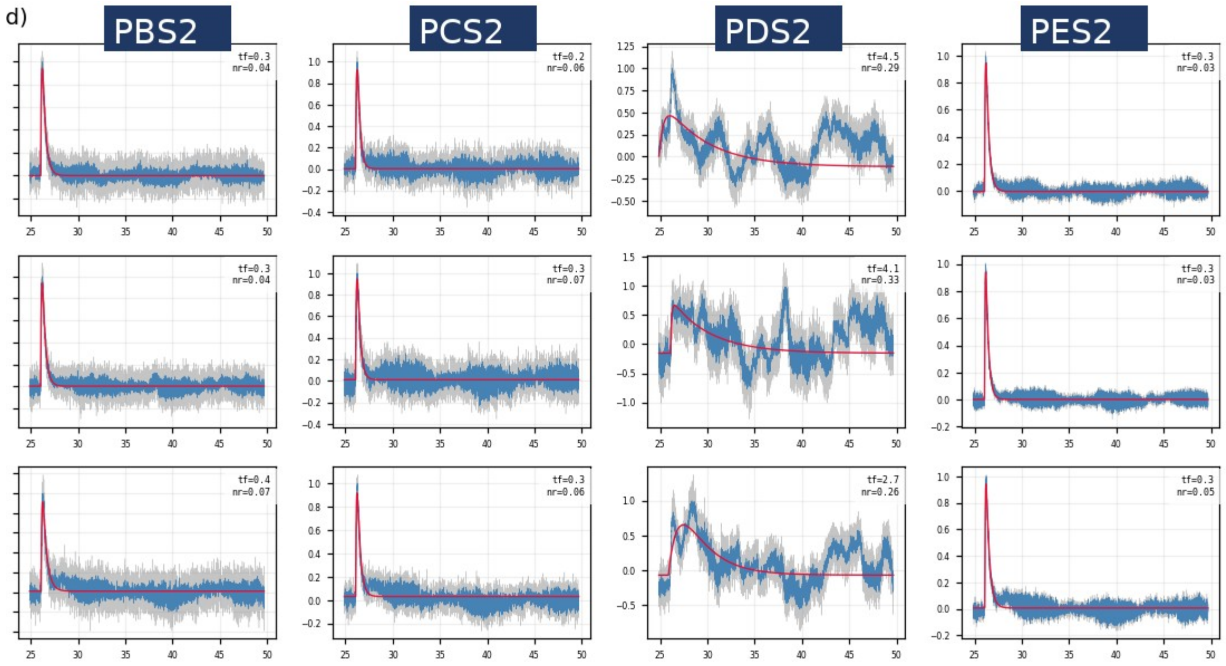
Z7: NRMSE-rejected events — raw traces are noise



Z7 PBS1 —— **passes the cut (kept)** —— **cut away = rejected**
faint blue = measured traces 1931 kept / 77 cut (~3.8%)

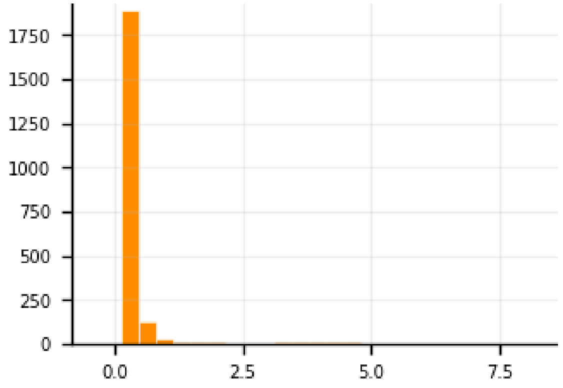
Follow-up 1 — the slow-fall tail is a one-channel artifact

- The post-cut fan still shows slow-fall tails → sample them: NRMSE ≤ 0.4 AND τ_fall > 1.5 ms, 10 random events, raw vs fit drawn in all 12 channels
- The sampled events are real pulses → kept. Their extreme fall times trace to one channel: only PDS2 swings (τ_fall median 0.51 ms vs ≈ 0.25 ms elsewhere) — a low-frequency disturbance
- Conclusion: no τ_fall cut — slow-fall events passing NRMSE stay in; the extreme tail is a one-channel artifact, not a slow pulse
- PDS2 alone: can not make useful templates; hard to fix in analysis/software, the noise itself needs to be fixed
- Temporary solution: use the PDS1 template in place of PDS2 (applied in the delivered ROOT files)

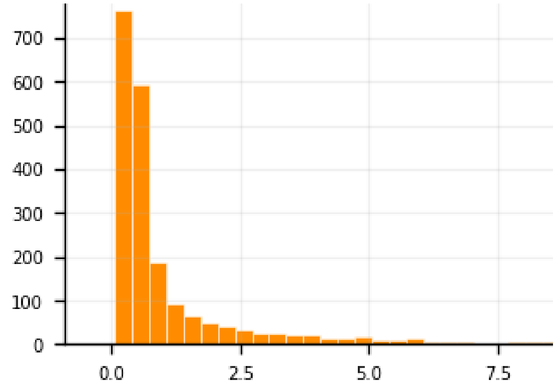


Z7, 3 sampled slow-fall events: normal in PBS2/PCS2/PES2, swings only in PDS2

Fitted τ_fall distribution, same 0–7 ms axis:



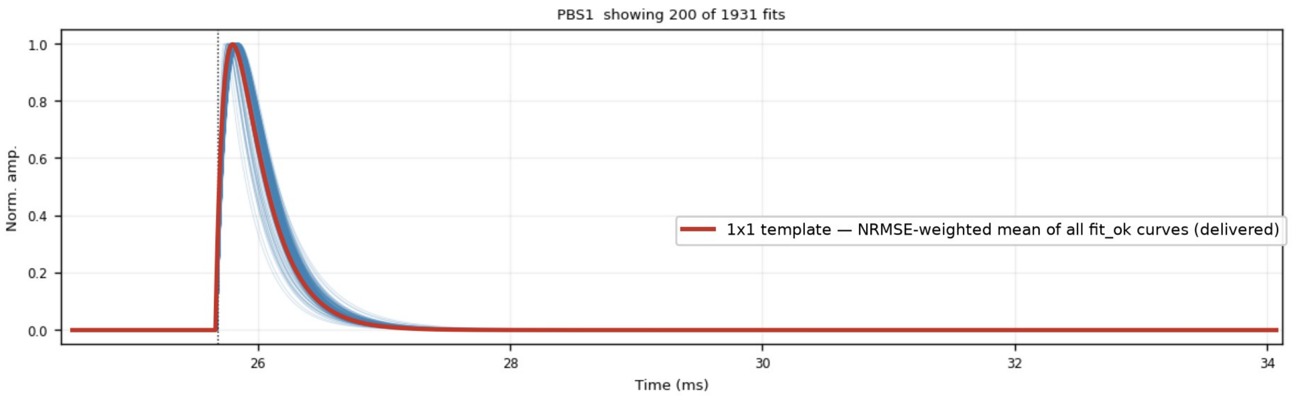
Z7 PAS1 — normal channel: narrow (median 0.25 ms)



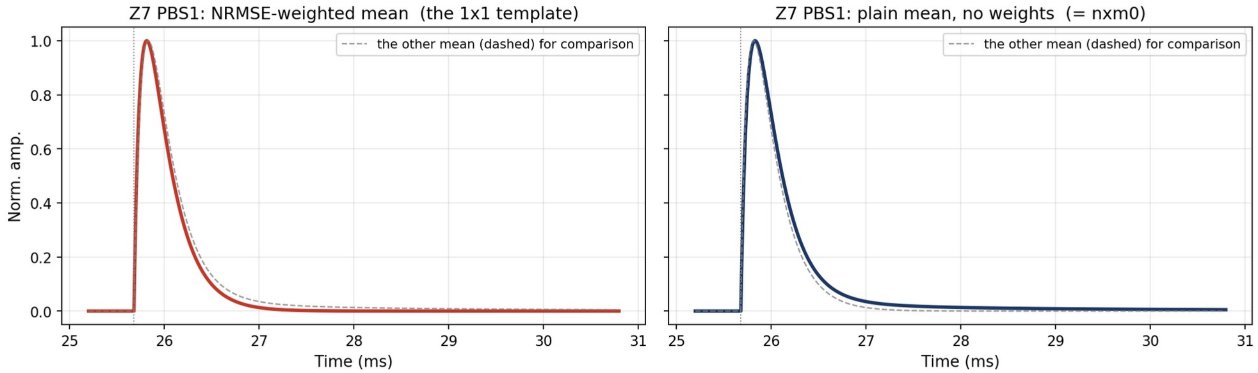
Z7 PDS2 — bad channel: broad tail (median 0.51 ms)

Template family 1 — analytic 2-exp, NRMSE-weighted (1x1)

- NRMSE-weighted mean of the fit_ok 2-exp curves
- Smooth & noise-free by construction
- ROOT histograms, peak-normalized (+ summed PT / PS1 / PS2)



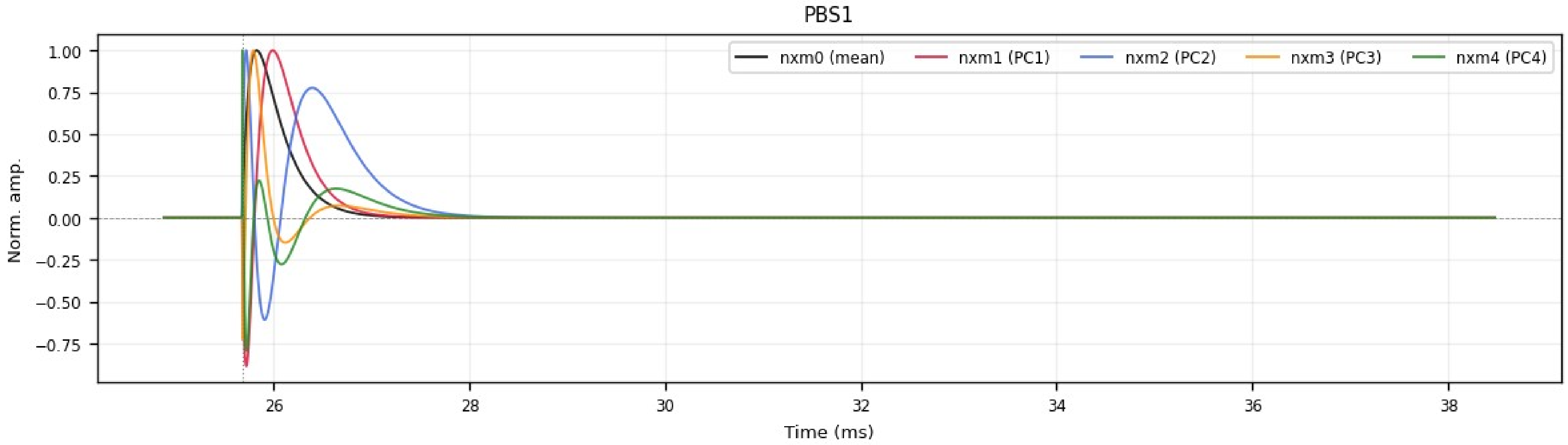
Z7 PBS1 — blue: the fitted curves (200 of 1931 drawn); red: the 1x1 template



Same curves, two averages (zoomed): NRMSE-weighted (the 1x1 template) vs plain mean (= nxm0); dashed gray = the other one

Template family 2 — NxM PCA templates

- Per-channel PCA over the fitted curves (fit_ok + NRMSE ≤ 0.4)
- nxm0 = mean shape; nxm1–4 = principal components; real pulse = Σ_i $amp_i \cdot nxm_i$
- PC1 + PC2 ≈ 96–98% of the shape variance; delivered peak-normalized



Z7 PBS1: nxm0 (mean) + nxm1–4 (PCs) — the oscillating components encode rise/fall-time variation

Template file

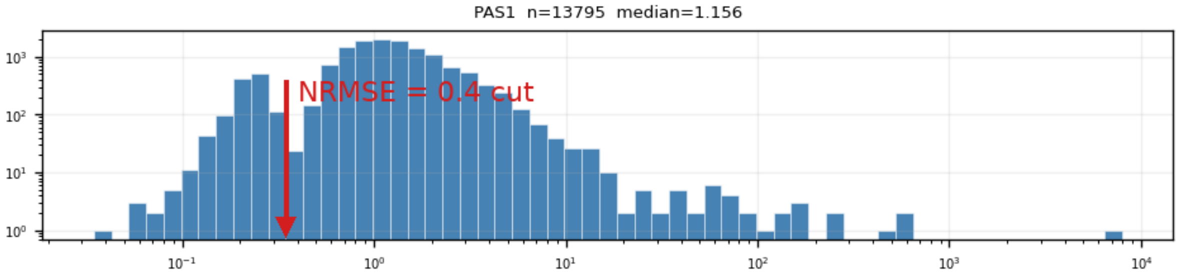
- Two template sets for all 13 detectors: 1x1 (2-exp weighted) + NxM PCA
- In cdmsbats_config feature branch, ready for merge

Future Steps

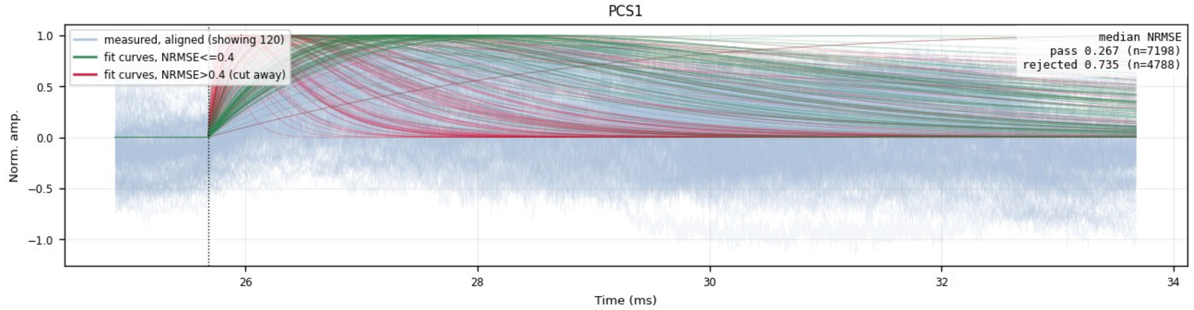
- Three-, four-exponential fits
- Exercise NxM processing
- Other detectors: worse noise, a few more cuts added, in the backup slides

Backup — weak detectors (example: Z22)

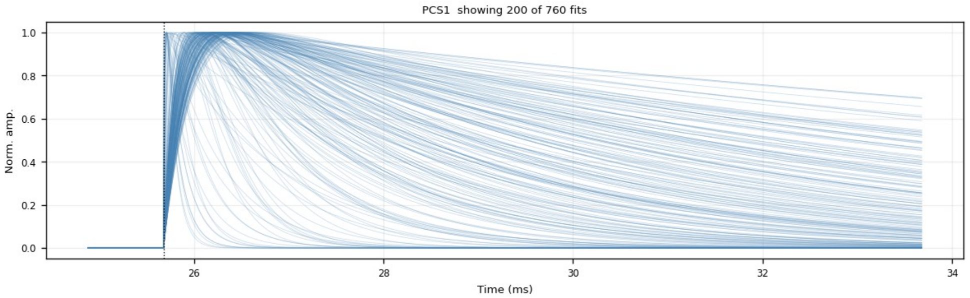
- Z1 / Z4 / Z6 / Z18 / Z19 / Z22 / Z24: K-line sits inside the noise population — the PTOF window admits a noise-dominated mixture
- Same bimodal NRMSE picture, noise peak dominates; the same 0.4 cut digs the real pulses out — Z22 PCS1: 7198 kept / 4788 cut (~40%)
- Kept bundle broader than on quiet detectors; steep red curves = fits latching onto sharp noise spikes, not fast pulses being cut
- τ_rise ≤ 0.3 ms (after NRMSE): removes 55–74% on weak zips (Z22: 71%) vs 1.6% on Z7 — residual slow drift



Z22 PAS1: NRMSE histogram — noise dominates



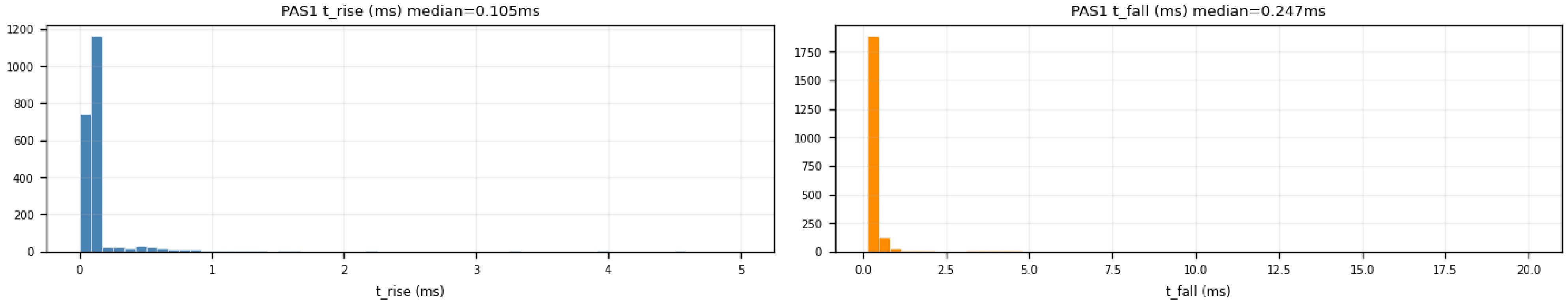
Z22 PCS1: kept (green) vs cut (red)



Z22 PCS1 after NRMSE ≤ 0.4 + τ_rise ≤ 0.3 ms — final shape family (760 fits)

Backup — the τ_rise ≤ 0.3 ms ceiling (PCA input)

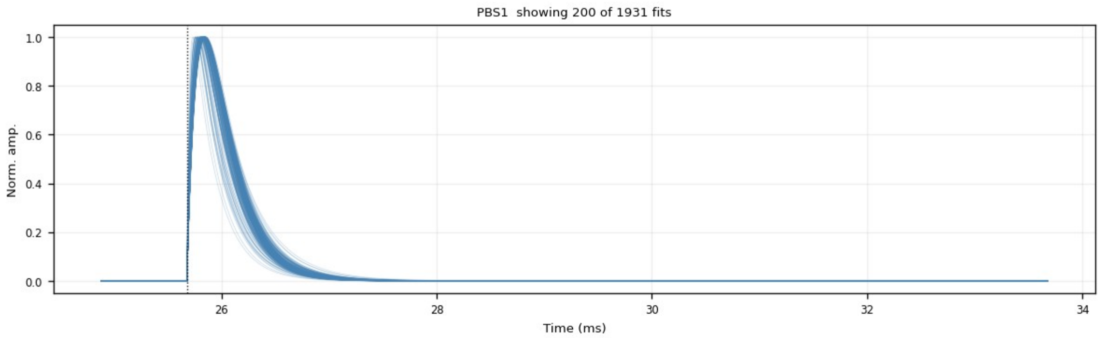
- Slow baseline drift survives NRMSE (a slow 2-exp hugs it) → mimics a slow rise
- Fast pulses at τ_rise ≈ 0.1 ms (p90 ≈ 0.15 ms) → ceiling at 0.3 ms blocks the drift tail
- Removed after the NRMSE cut: Z7 1.6% (quiet 2–6%) — weak 55–74% (Z22 71%) — all zips 54%
- Trade-off: trims the very slowest genuine pulses → decision pending



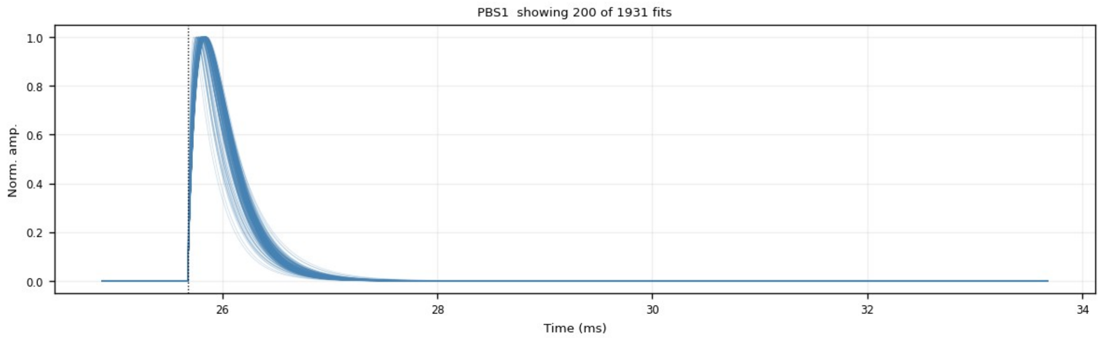
Z7 PAS1: fitted τ_rise (median 0.105 ms) and τ_fall distributions

Backup — τ_rise cut: a good vs a bad channel (Z7)

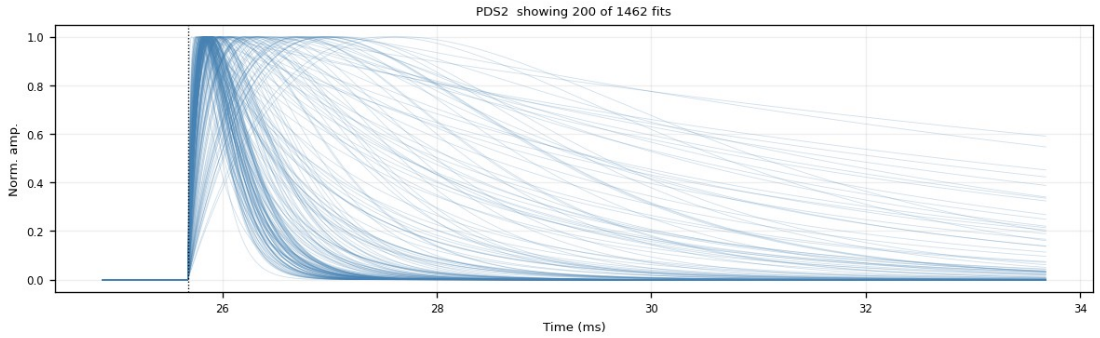
- Good channel PBS1: tight clean bundle — the τ_rise ≤ 0.3 ms cut removes ~0% (nearly free on quiet channels)
- Bad channel PDS2: broad, messy fan — the channel itself is bad; the cut trims the slow-rise curves, 1462 → 1154 (21%)



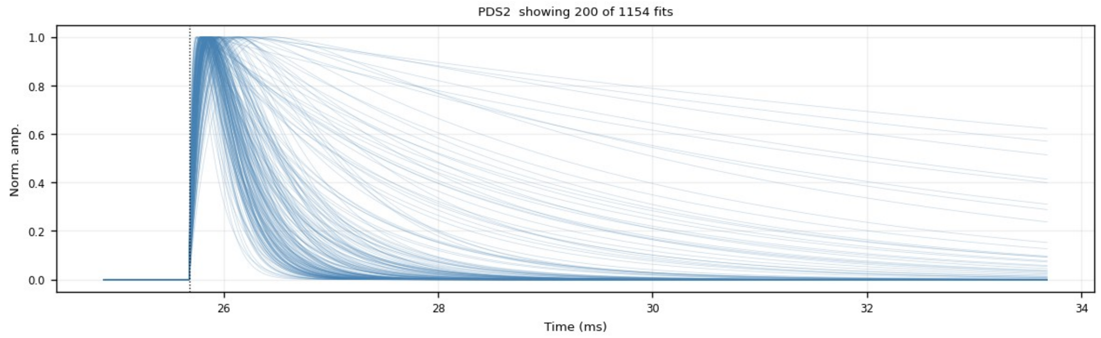
PBS1 (good) — NRMSE ≤ 0.4



PBS1 (good) — + τ_rise ≤ 0.3 ms (unchanged)



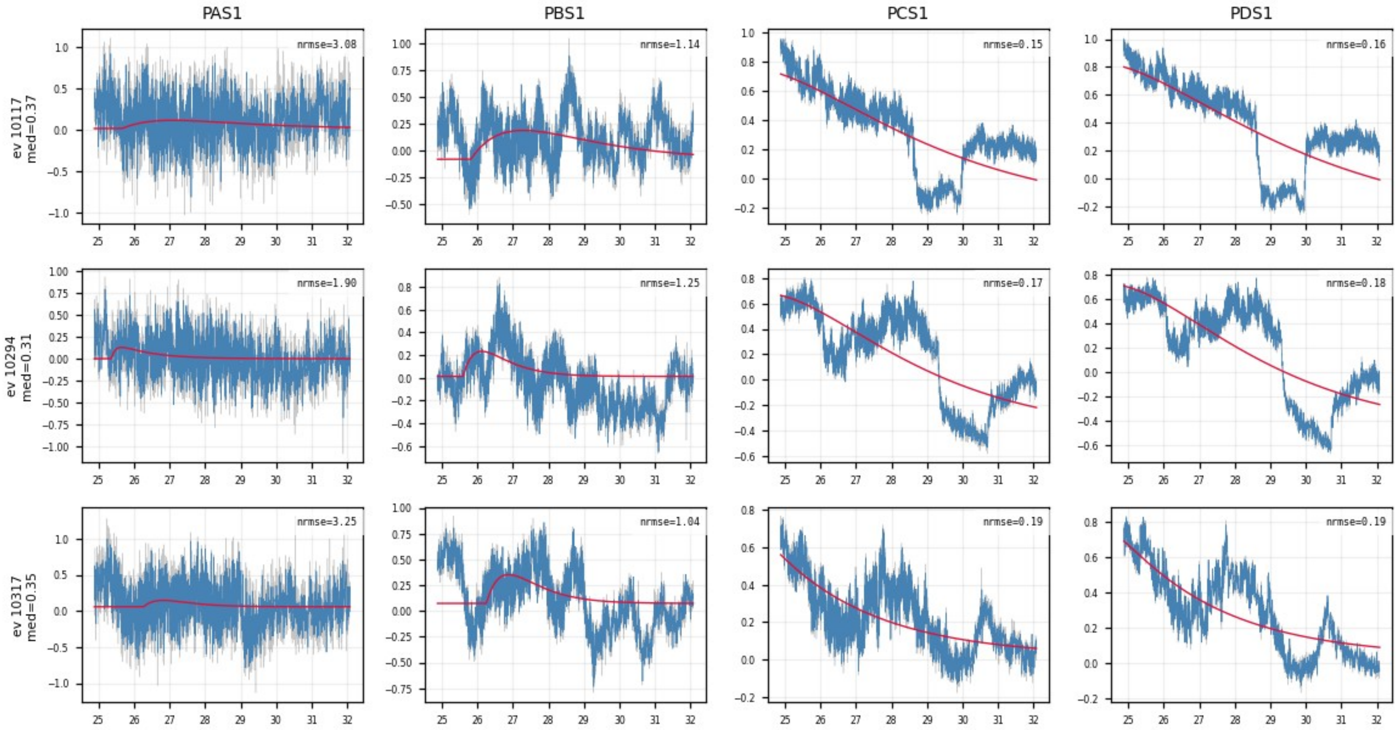
PDS2 (bad) — NRMSE ≤ 0.4



PDS2 (bad) — + τ_rise ≤ 0.3 ms (slow risers trimmed)

Backup — what the τ_rise cut removes: noise NRMSE missed (Z22)

- Exactly the events that PASS NRMSE ≤ 0.4 but are REMOVED by τ_rise ≤ 0.3 ms — the noise 0.4 let through, 0.3 catches
- PCS1 / PDS1 pass at NRMSE ≈ 0.15 (a slow 2-exp hugs a slow baseline drift) — but the raw trace has no clean fast pulse
- Trade-off: the same cut also trims genuine slow-rise pulses — documented, decision pending

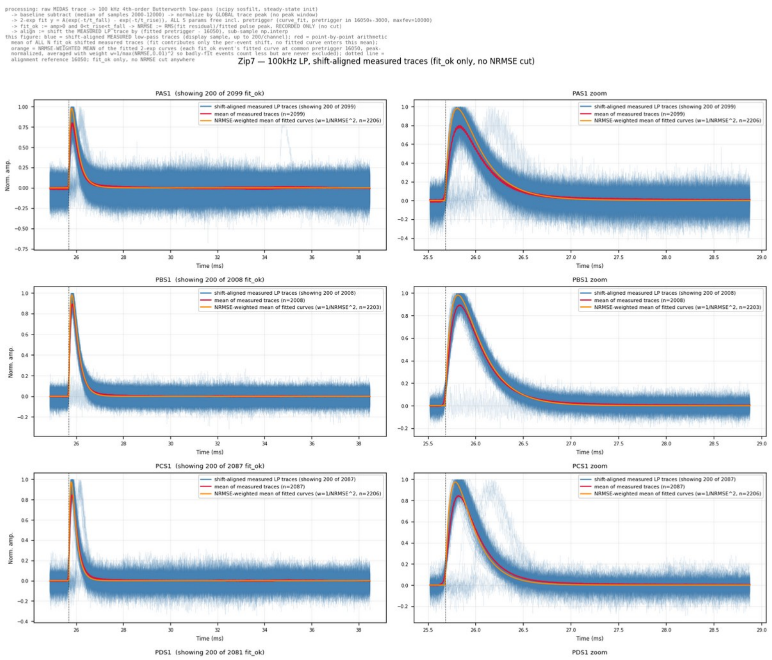


Z22: 3 events × PAS1/PBS1/PCS1/PDS1 — pass NRMSE ≤ 0.4, removed by τ_rise ≤ 0.3 ms; PCS1/PDS1 are pure slow drift

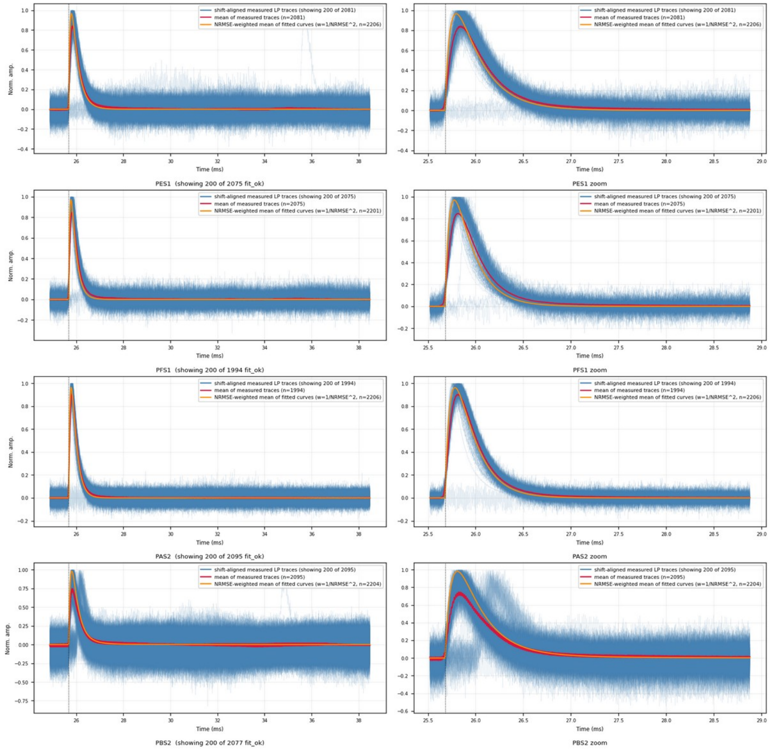
Backup — Z7 · aligned_overlay

shift-aligned measured traces · figure: zip7_lp_aligned_overlay.png

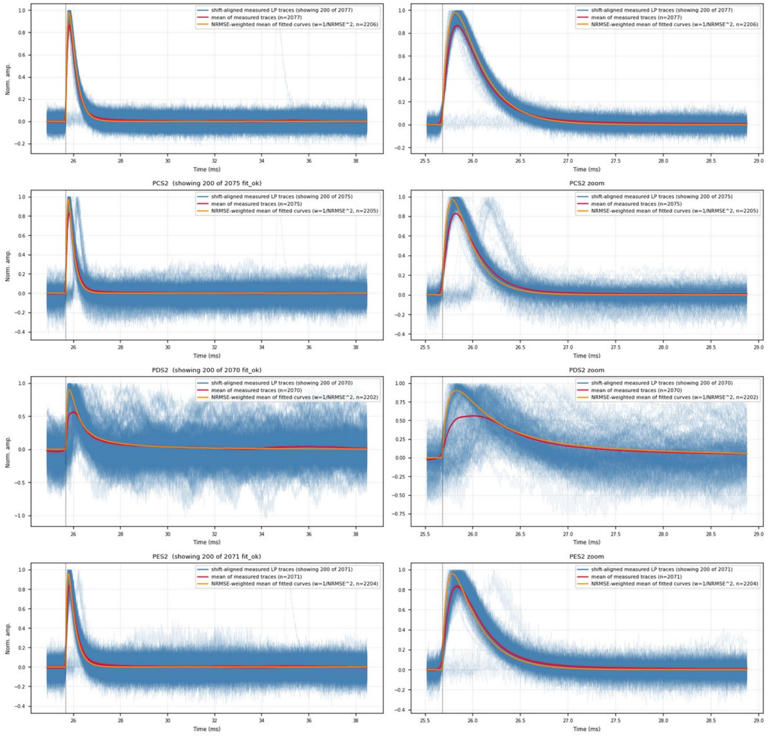
Blue = measured LP traces shifted by (fitted pretrigger − 16050); red = point-by-point mean of ALL fit_ok events; orange = NRMSE-weighted mean of the fitted curves (w = 1/max(NRMSE,0.01)²). Rise edges line up at 16050.



part 1/3 (top to bottom)



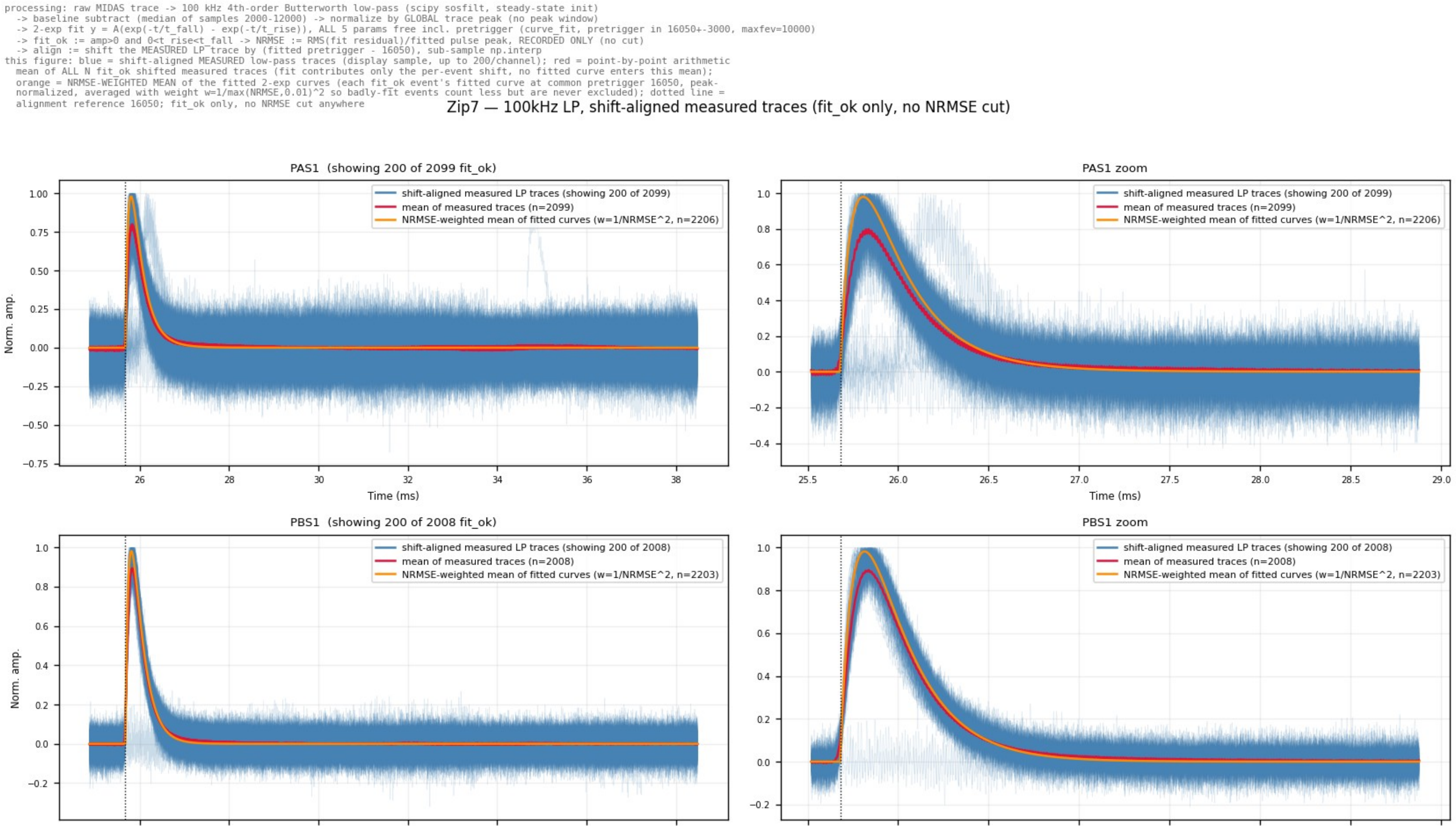
part 2/3 (top to bottom)



part 3/3 (top to bottom)

Backup — Z7 · aligned_overlay (detail 1:1)

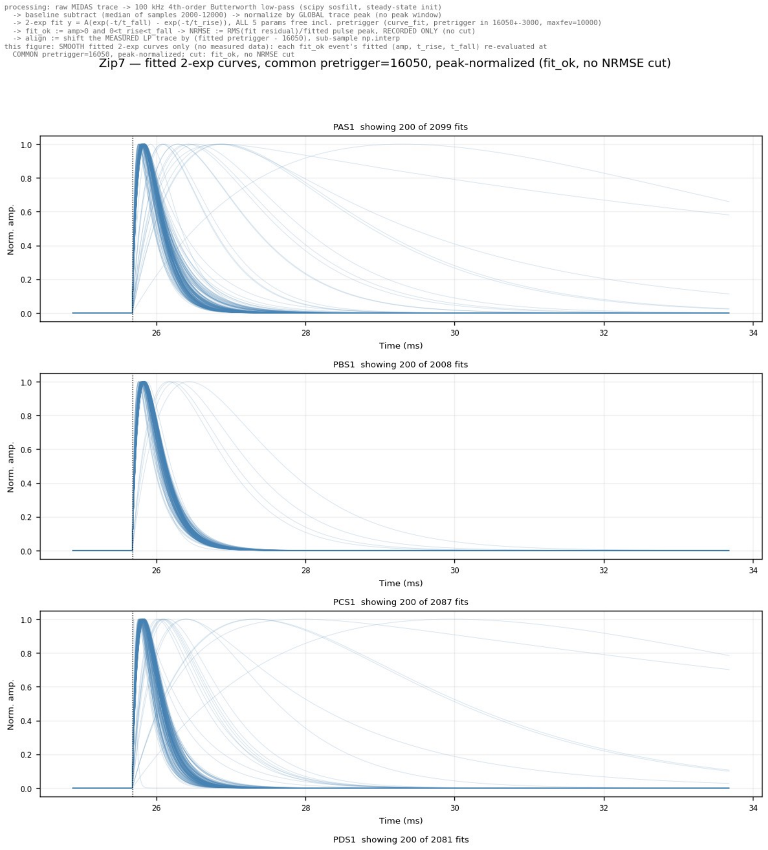
top region of the same figure at full resolution — whole figure on the previous slide



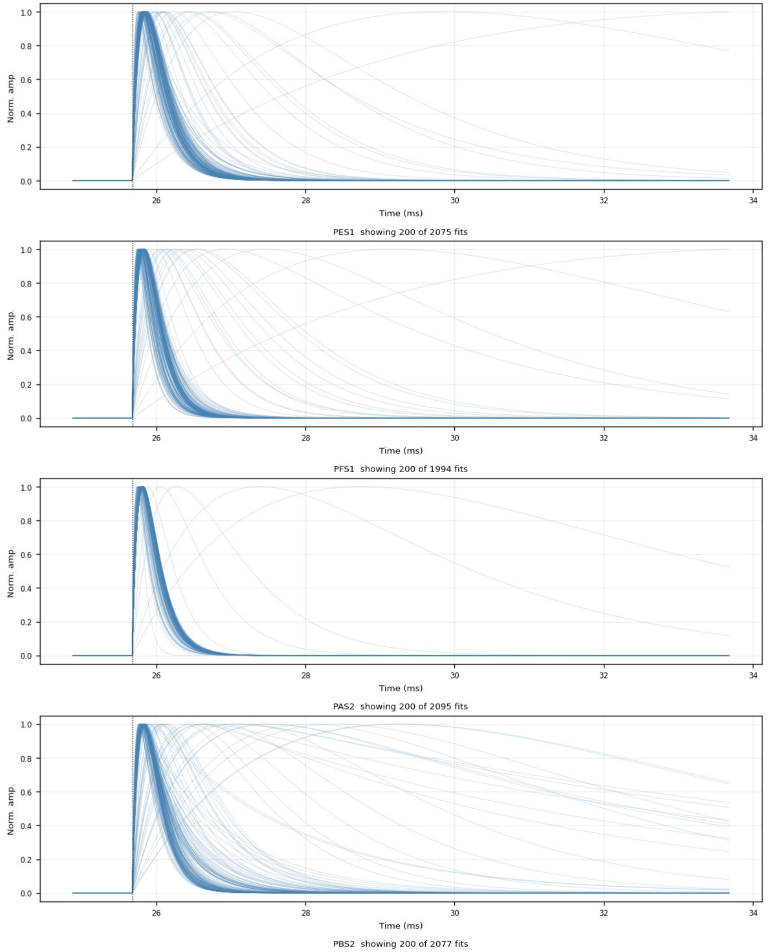
Backup — Z7 · fitted_curves_overlay

fan of fitted 2-exp curves (no cut) · figure: zip7_fitted_curves_overlay.png

Every fit_ok event's fitted curve re-evaluated at common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.



part 1/3 (top to bottom)



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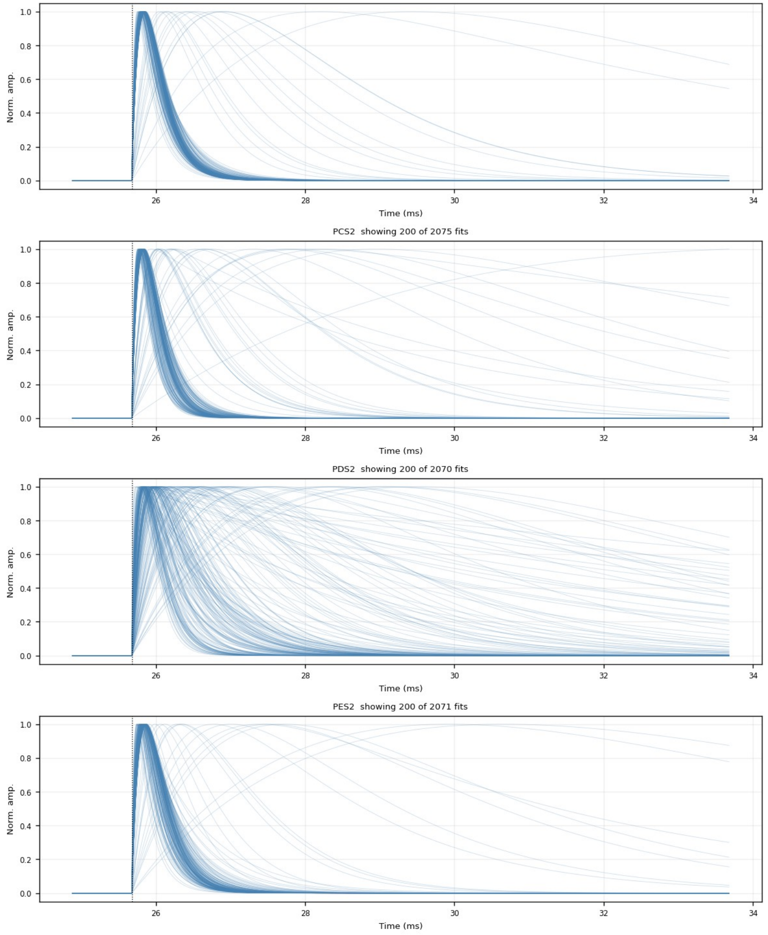

part 3/3 (top to bottom)

Backup — Z7 · fitted_curves_overlay (detail 1:1)

top region of the same figure at full resolution — whole figure on the previous slide

```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
  -> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
  -> 2-exp fit y = A(exp(-t/t_fall) - exp(-t/t_rise)), ALL 5 params free incl. pretrigger (curve_fit, pretrigger in 16050+-3000, maxfev=10000)
  -> fit_ok := amp>0 and 0<t_rise<t_fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
  -> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: SMOOTH fitted 2-exp curves only (no measured data): each fit_ok event's fitted (amp, t_rise, t_fall) re-evaluated at
  COMMON pretrigger=16050, peak-normalized; cut: fit_ok, no NRMSE cut
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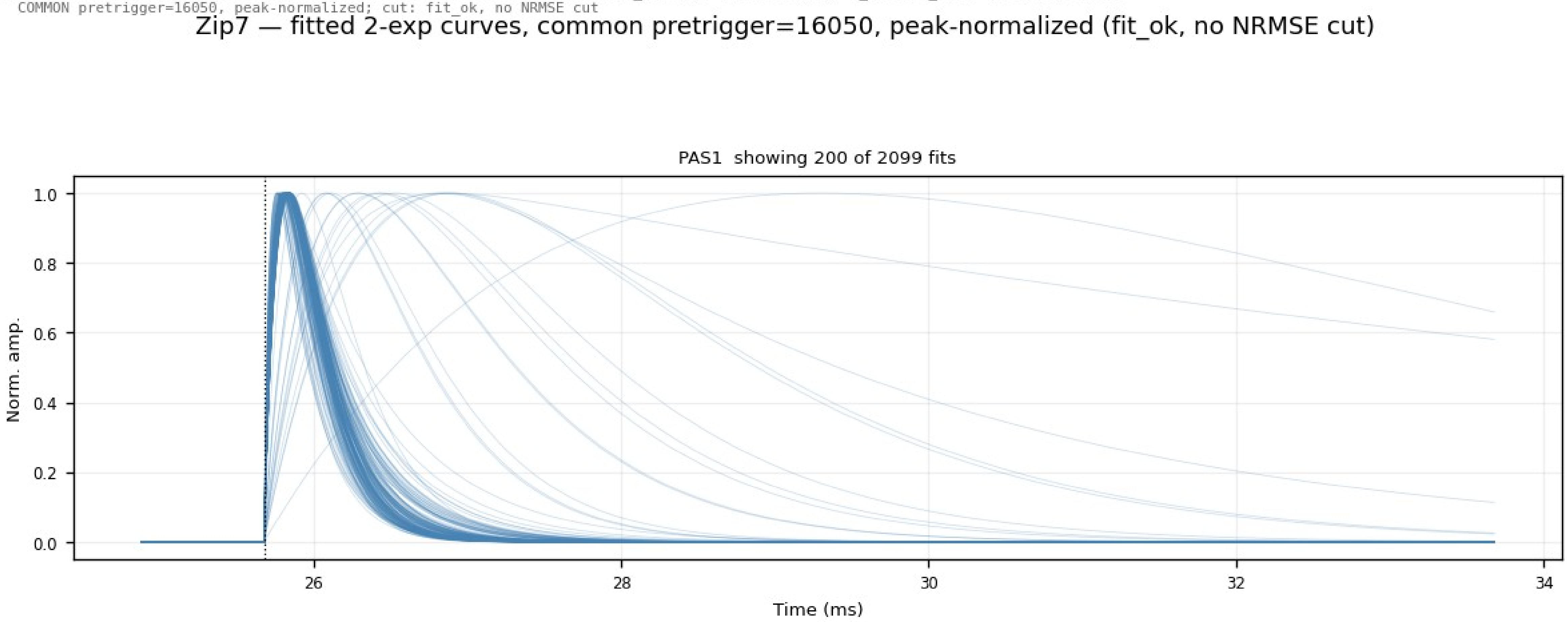
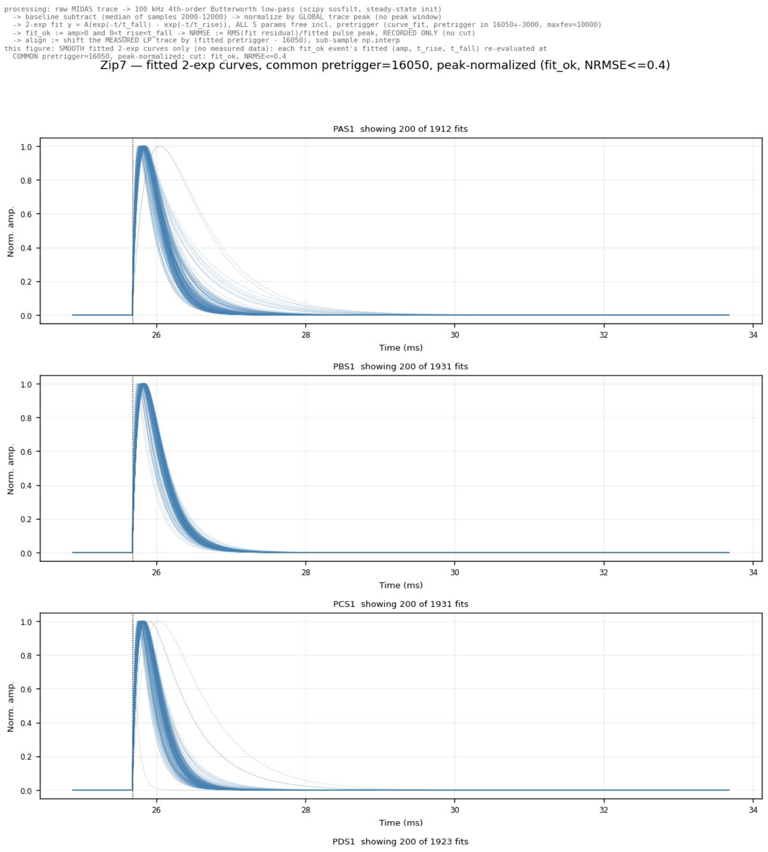


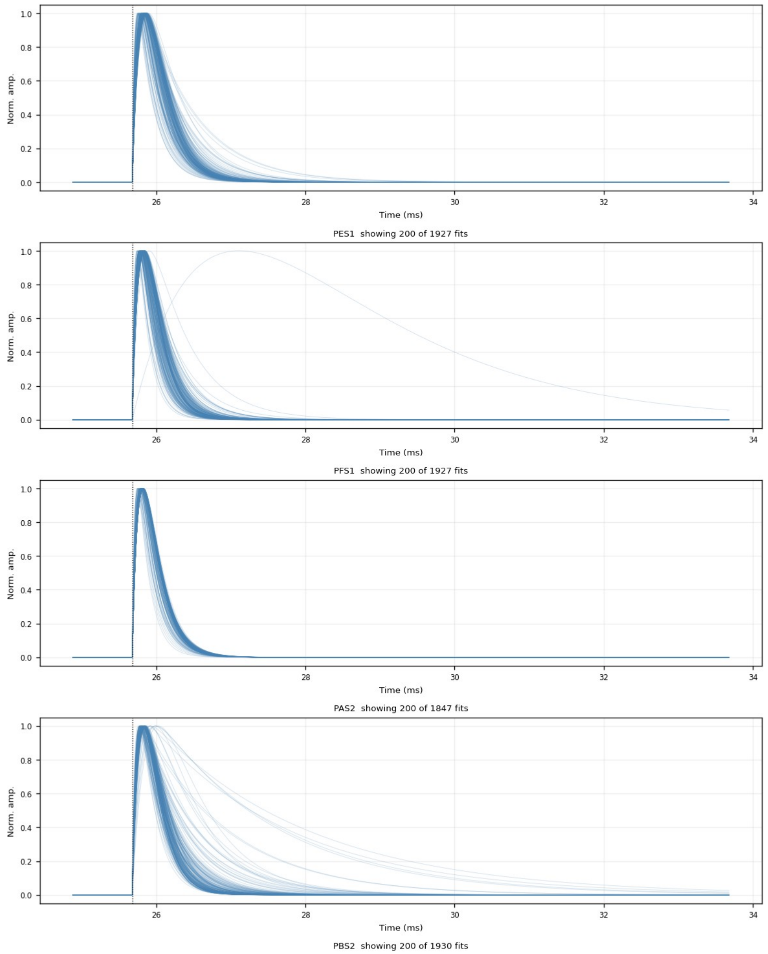
Backup — Z7 · fitted_curves_overlay

after NRMSE ≤ 0.4 · figure: zip7_fitted_curves_overlay_nrmse0.4.png

Same fan after the NRMSE cut: the noise fan collapses; the fast-pulse bundle remains.



part 1/3 (top to bottom)



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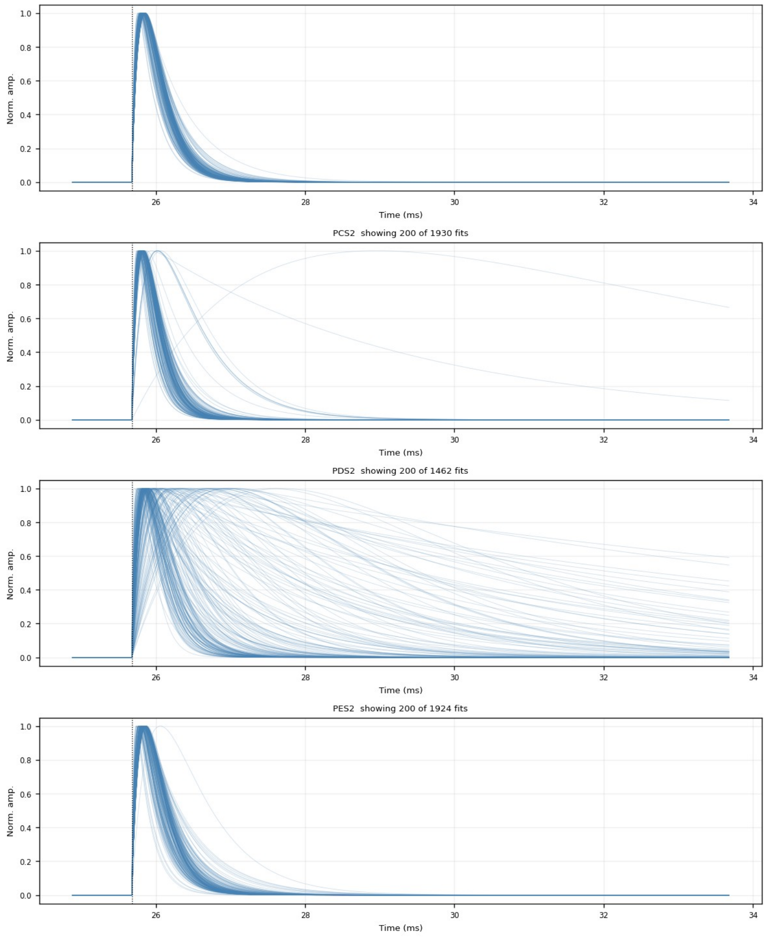

part 3/3 (top to bottom)

Backup — Z7 · fitted_curves_overlay (detail 1:1)

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  COMMON pretrigger=16050, peak-normalized; cut: fit_ok, NRMSE<=0.4
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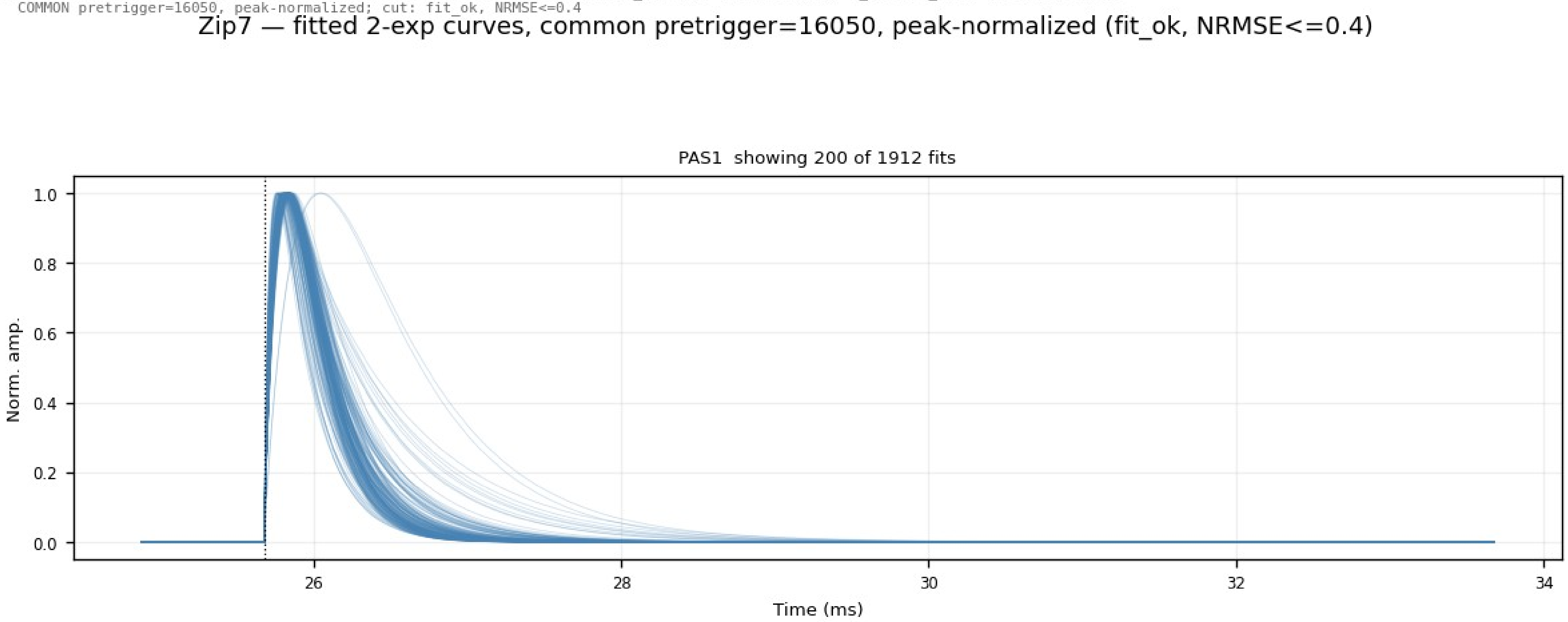
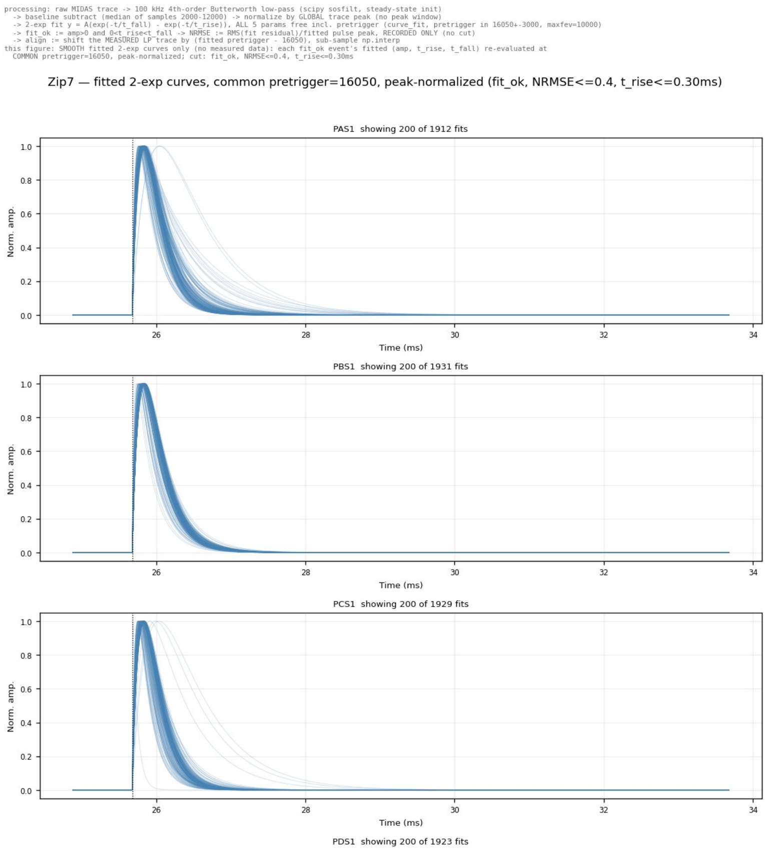


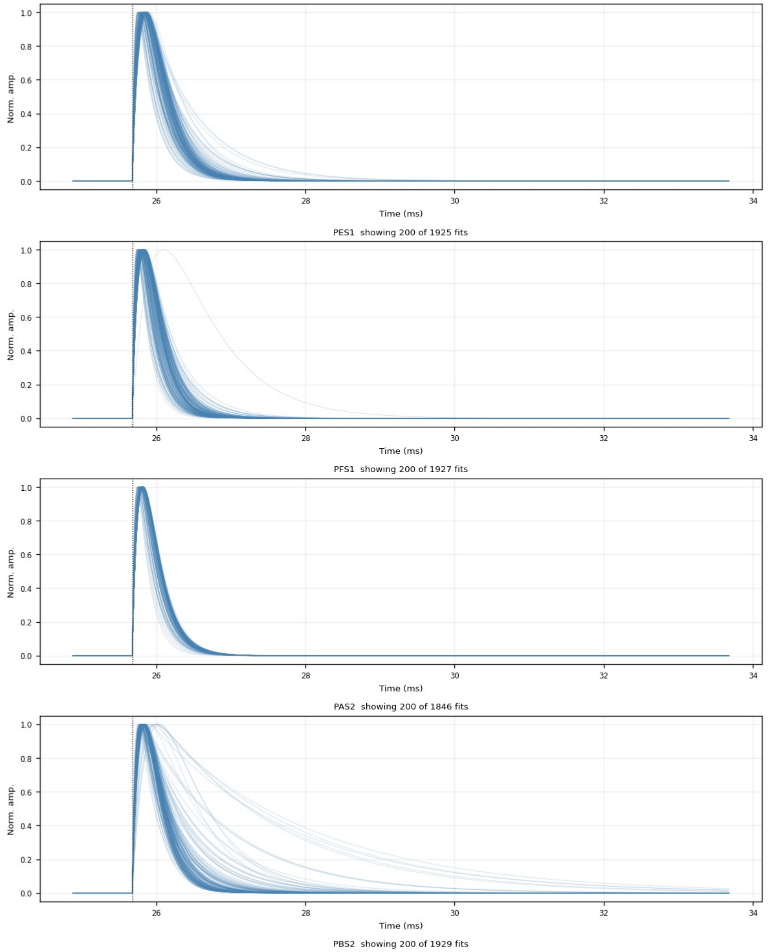
Backup — Z7 · fitted_curves_overlay

after NRMSE ≤ 0.4 and τ_rise ≤ 0.3 ms · figure: zip7_fitted_curves_overlay_nrmse0.4_trise0.30ms.png

The exact PCA input population: clean fast-pulse bundle only.



part 1/3 (top to bottom)



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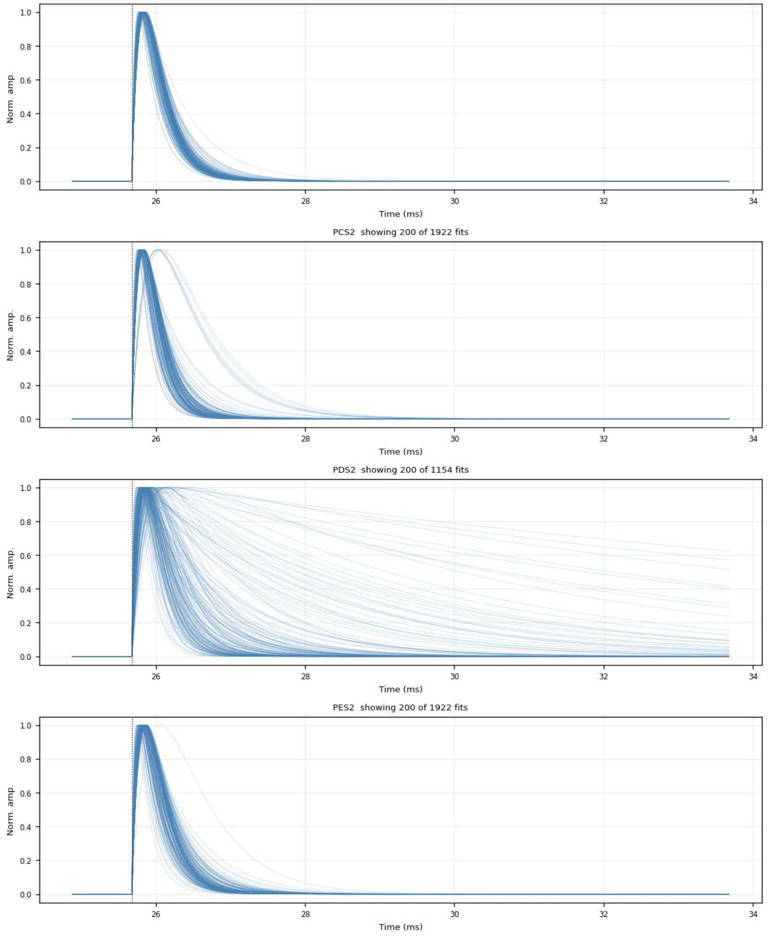

part 3/3 (top to bottom)

Backup — Z7 · fitted_curves_overlay (detail 1:1)

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  COMMON pretrigger=16050, peak-normalized; cut: fit_ok, NRMSE<=0.4, t_rise<=0.30ms
```
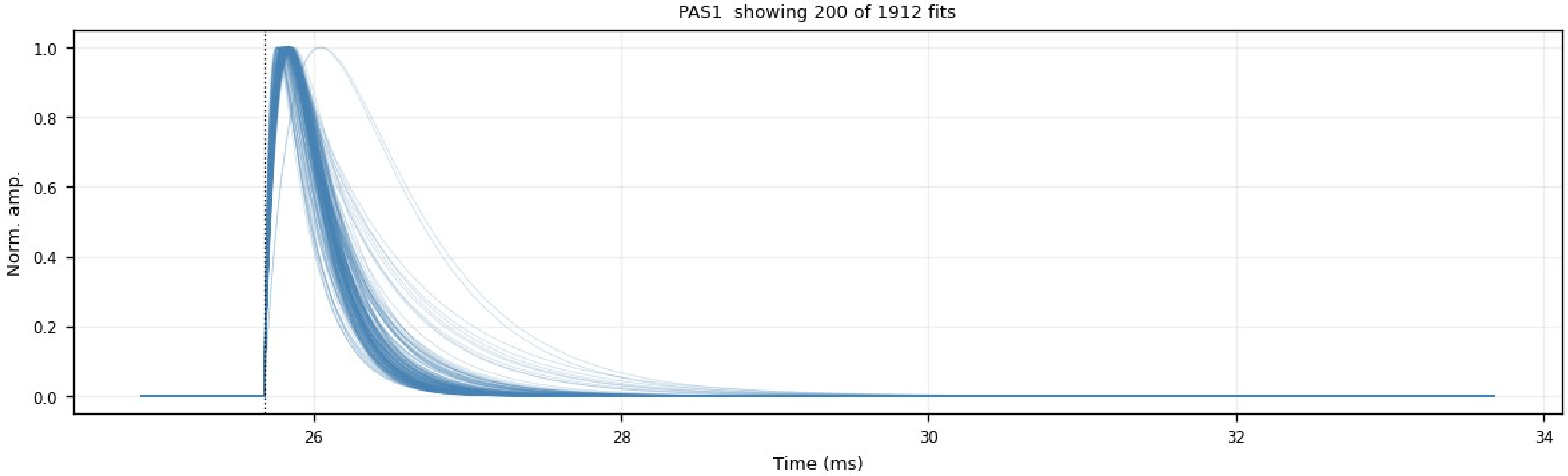
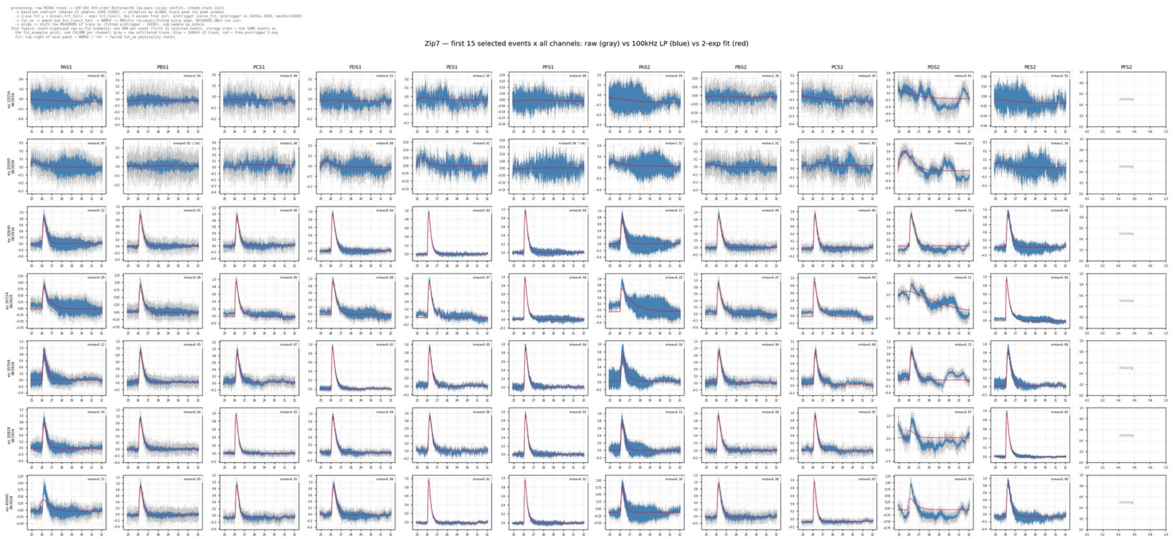

Zip7 — fitted 2-exp curves, common pretrigger=16050, peak-normalized (fit_ok, NRMSE<=0.4, t_rise<=0.30ms)



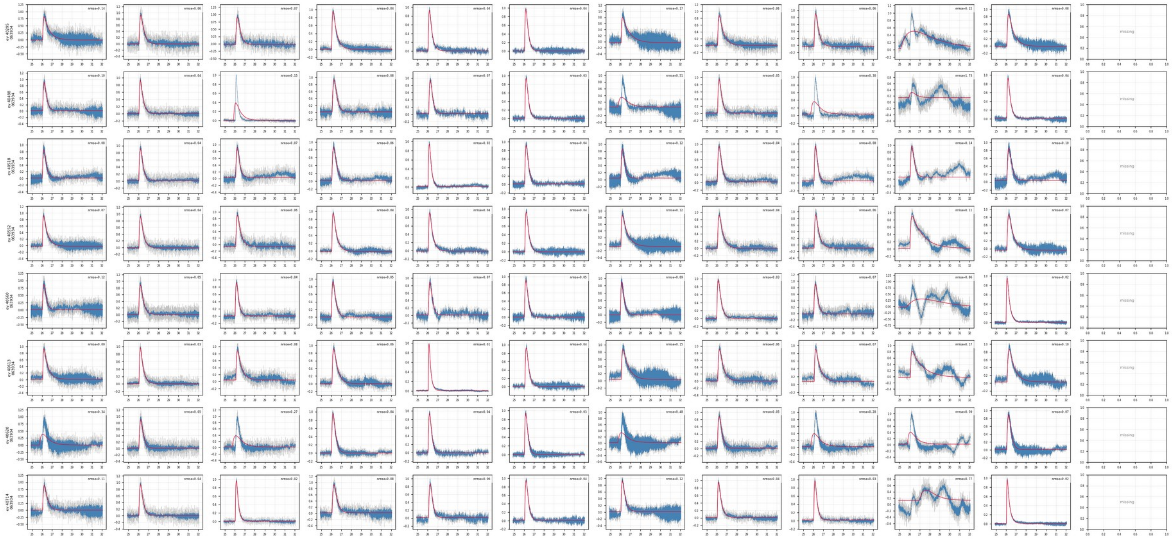
Backup — Z7 · raw_vs_fit_examples

first 15 events × all channels · figure: zip7_raw_vs_fit_examples.png

Gray = raw unfiltered, blue = 100 kHz LP, red = 2-exp fit; per-panel NRMSE top-right. Same 15 events as fit_examples; noise vs pulse size is directly visible.



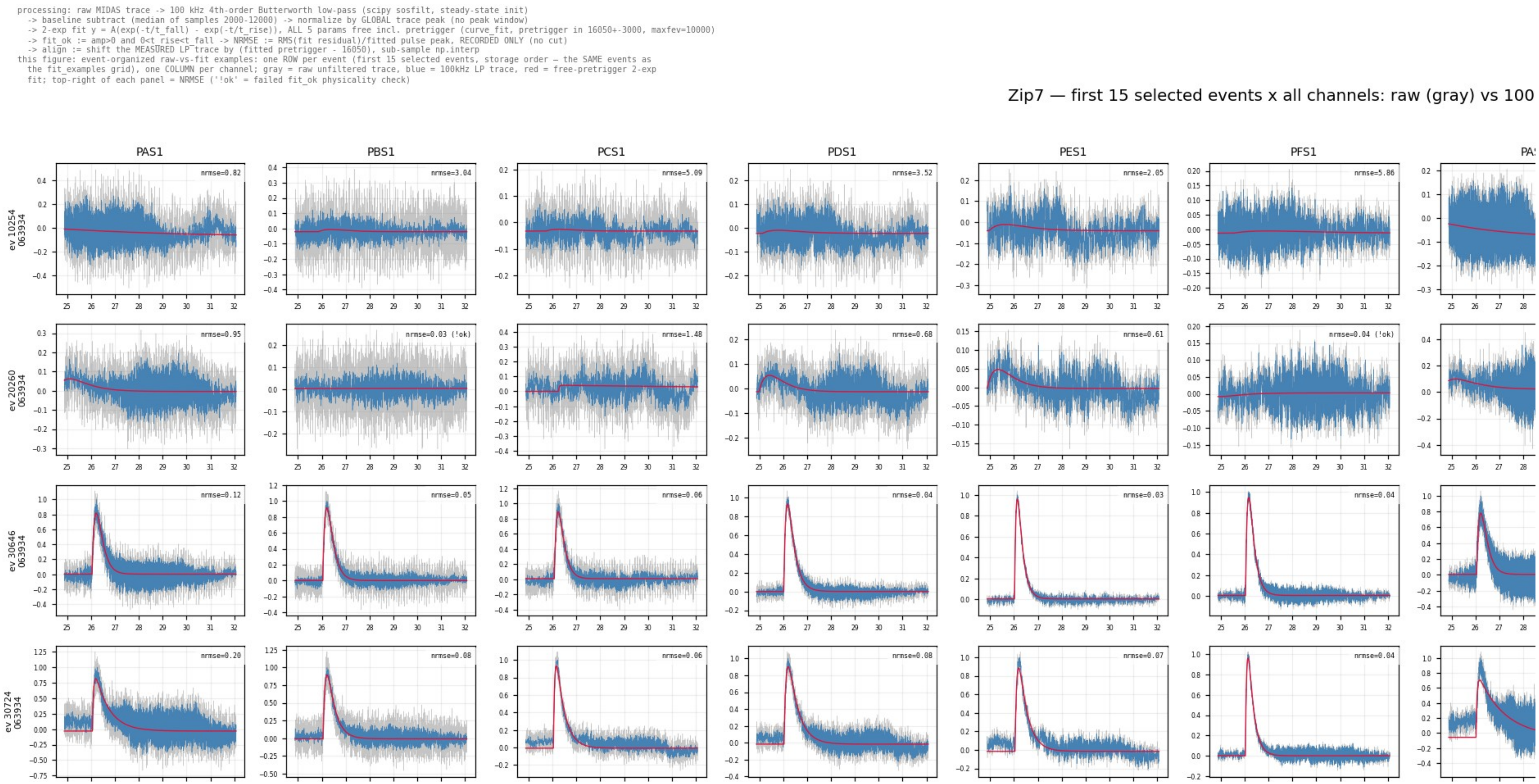
part 1/2 (top to bottom)



part 2/2 (top to bottom)

Backup — Z7 · raw_vs_fit_examples (detail 1:1)

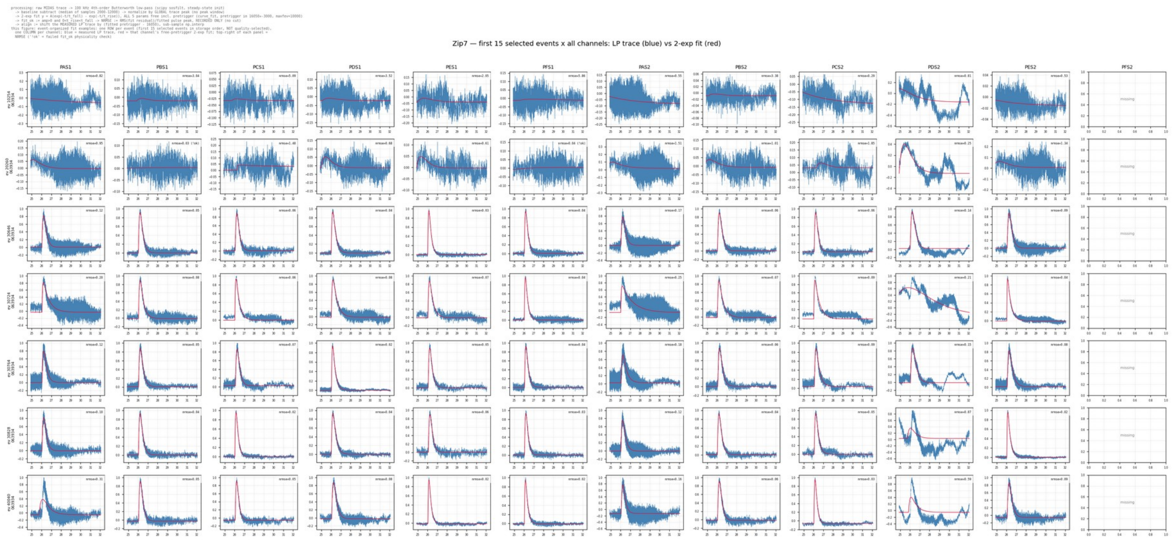
top region of the same figure at full resolution — whole figure on the previous slide



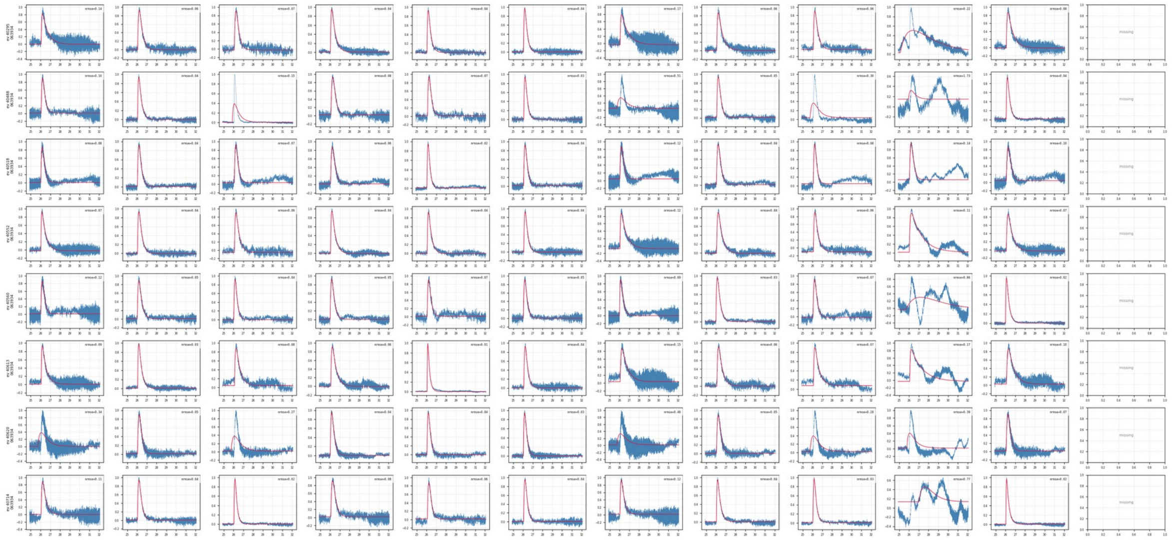
Backup — Z7 · fit_examples

first 15 events × all channels (LP vs fit) · figure: zip7_fit_examples.png

One ROW per event, one COLUMN per channel. A real event shows a clean pulse and a good fit in every channel simultaneously; a noise trigger fails across the whole row.



part 1/2 (top to bottom)

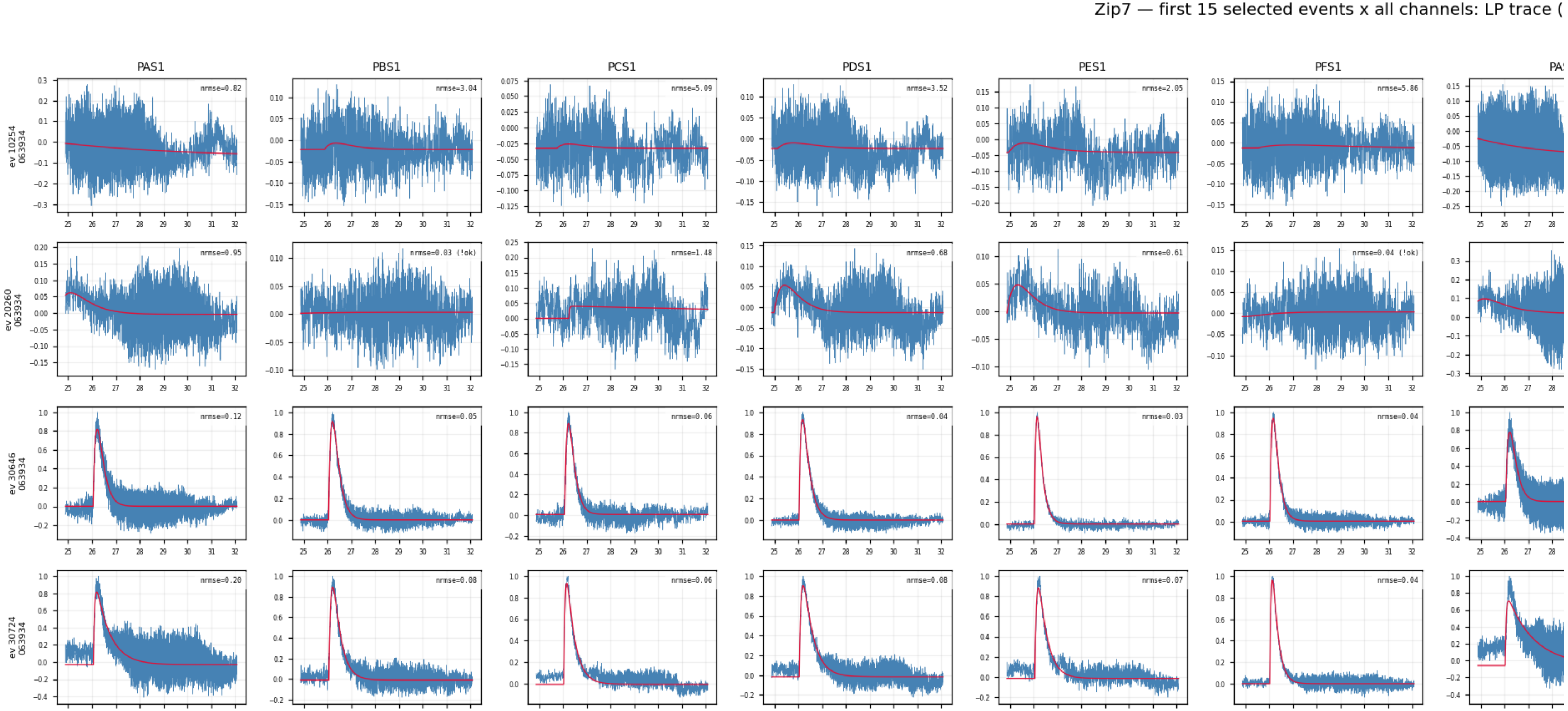


part 2/2 (top to bottom)

Backup — Z7 · fit_examples (detail 1:1)

top region of the same figure at full resolution — whole figure on the previous slide

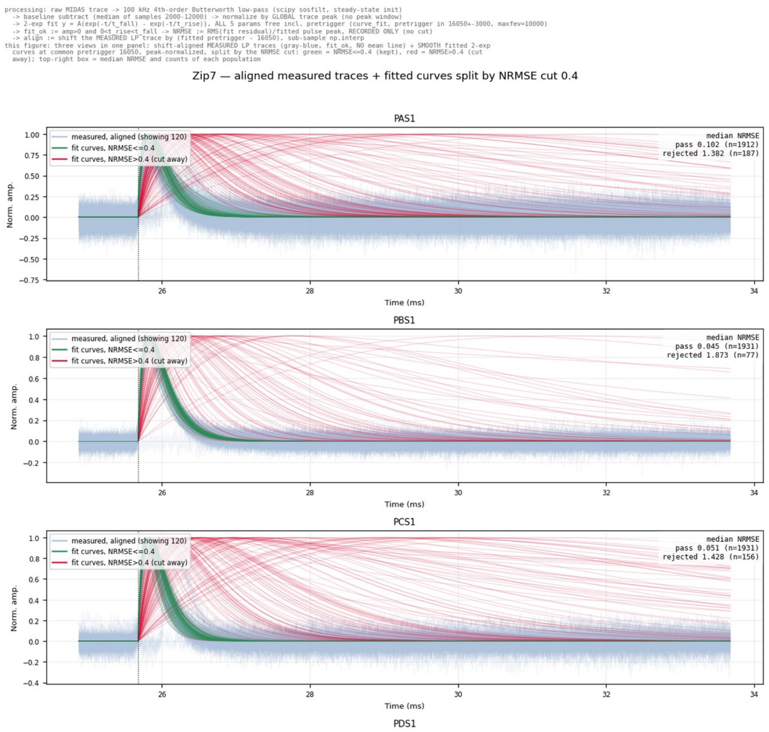
```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
  -> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
  -> 2-exp fit y = A(exp(-t/t_fall) - exp(-t/t_rise)), ALL 5 params free incl. pretrigger (curve_fit, pretrigger in 16050+-3000, maxfev=10000)
  -> fit_ok := amp>0 and 0<t_rise<t_fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
  -> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: event-organized fit examples: one ROW per event (first 15 selected events in storage order, NOT quality-selected),
  one COLUMN per channel; blue = measured LP trace, red = that channel's free-pretrigger 2-exp fit; top-right of each panel =
  NRMSE ('!ok' = failed fit_ok physicality check)
```



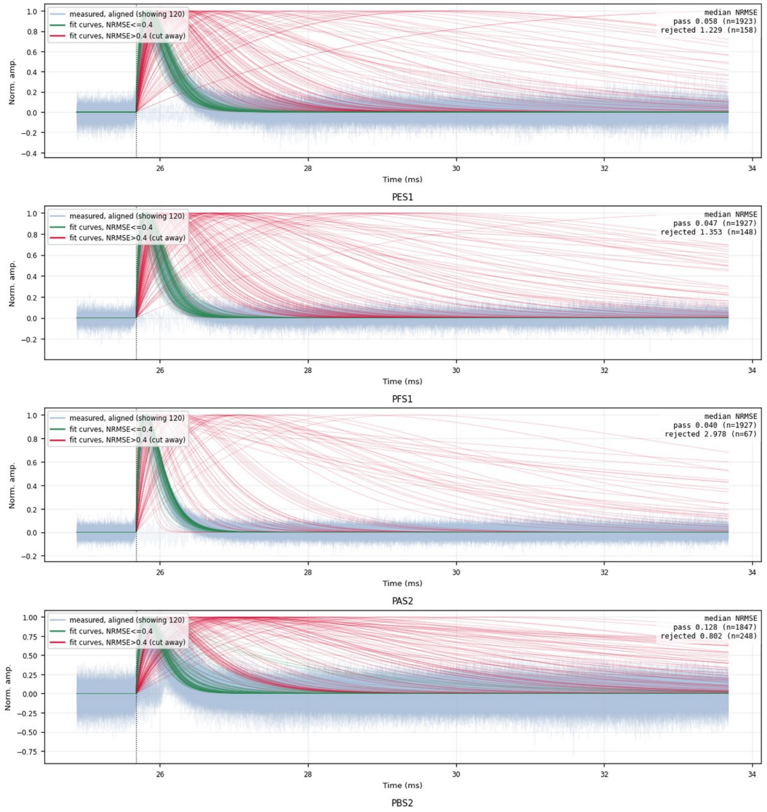
Backup — Z7 · overlay_fan_cut

measured traces + kept/rejected fits · figure: zip7_overlay_fan_cut_nrmse0.4.png

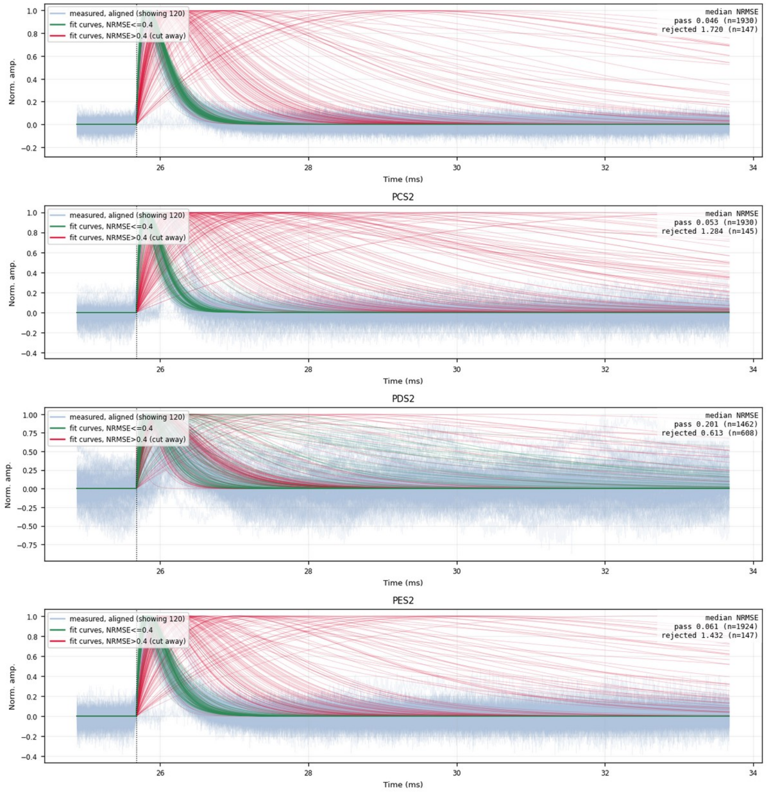
Gray-blue = aligned measured traces; green = fitted curves with NRMSE ≤ 0.4 (kept); red = NRMSE > 0.4 (cut away). Median NRMSE and counts of both populations stamped per channel.



part 1/3 (top to bottom)



part 2/3 (top to bottom)

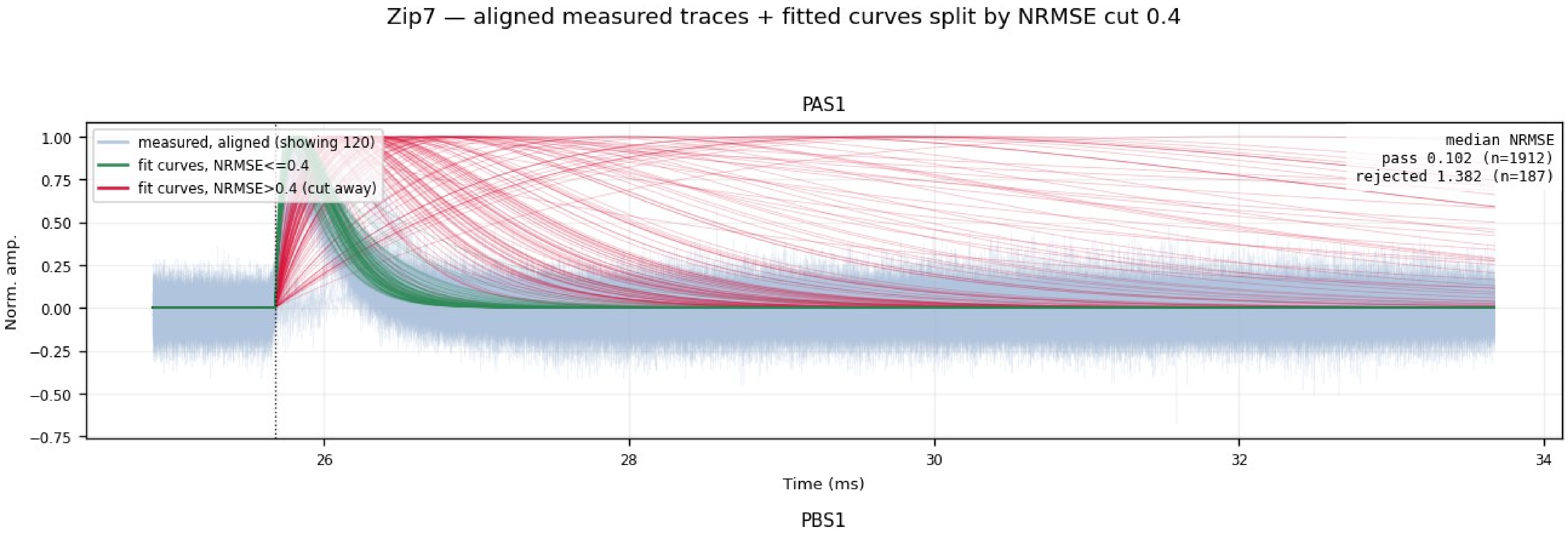


part 3/3 (top to bottom)

Backup — Z7 · overlay_fan_cut (detail 1:1)

top region of the same figure at full resolution — whole figure on the previous slide

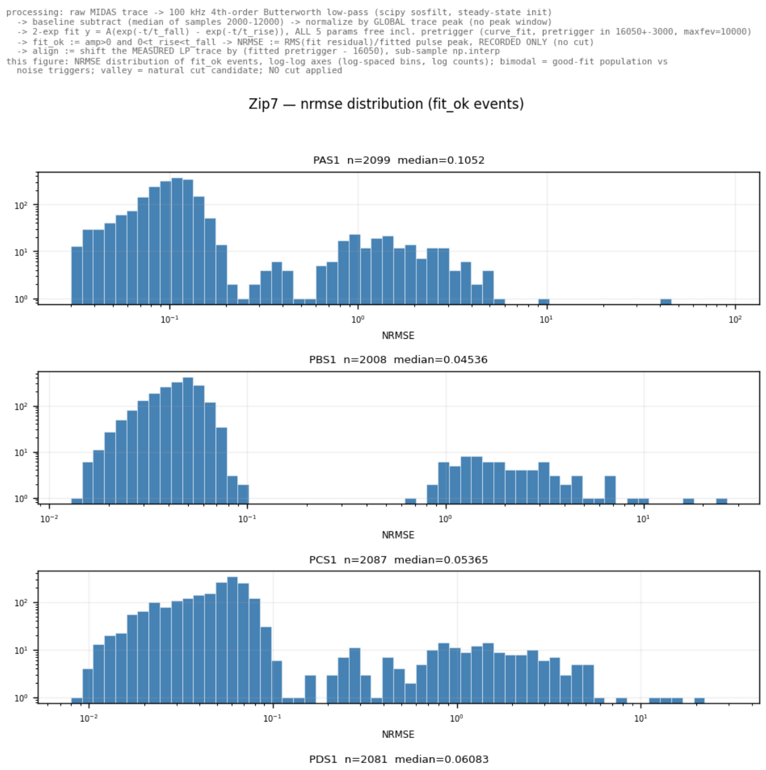
```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
  -> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
  -> 2-exp fit y = A(exp(-t/t_fall) - exp(-t/t_rise)), ALL 5 params free incl. pretrigger (curve_fit, pretrigger in 16050+-3000, maxfev=10000)
  -> fit_ok := amp>0 and 0<t_rise<t_fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
  -> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: three views in one panel: shift-aligned MEASURED LP traces (gray-blue, fit_ok, NO mean line) + SMOOTH fitted 2-exp
  curves at common pretrigger 16050, peak-normalized, split by the NRMSE cut: green = NRMSE<=0.4 (kept), red = NRMSE>0.4 (cut
  away); top-right box = median NRMSE and counts of each population
```



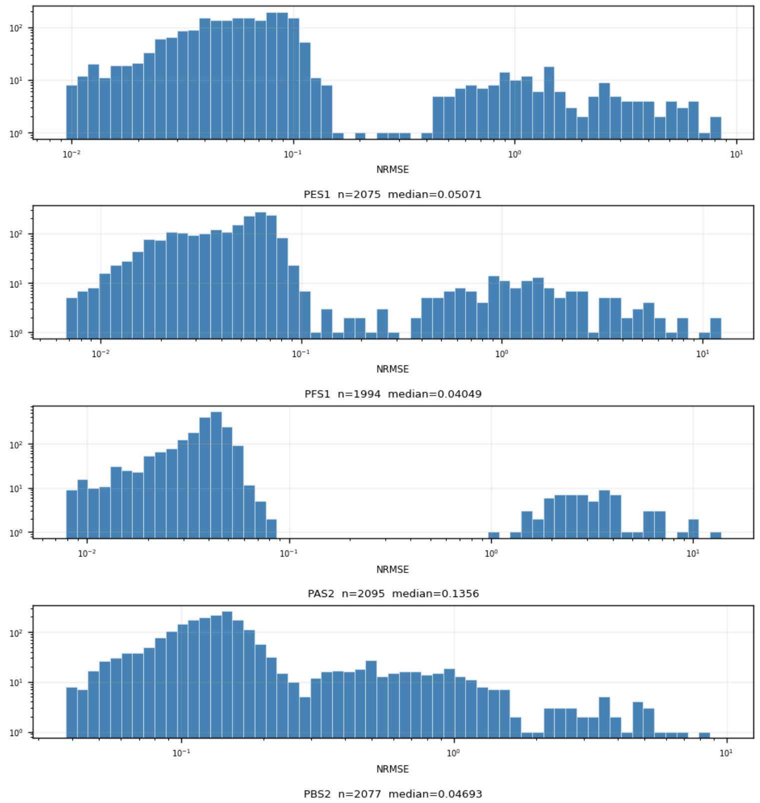
Backup — Z7 · nrmse

NRMSE distribution (log-log) · figure: zip7_nrmse.png

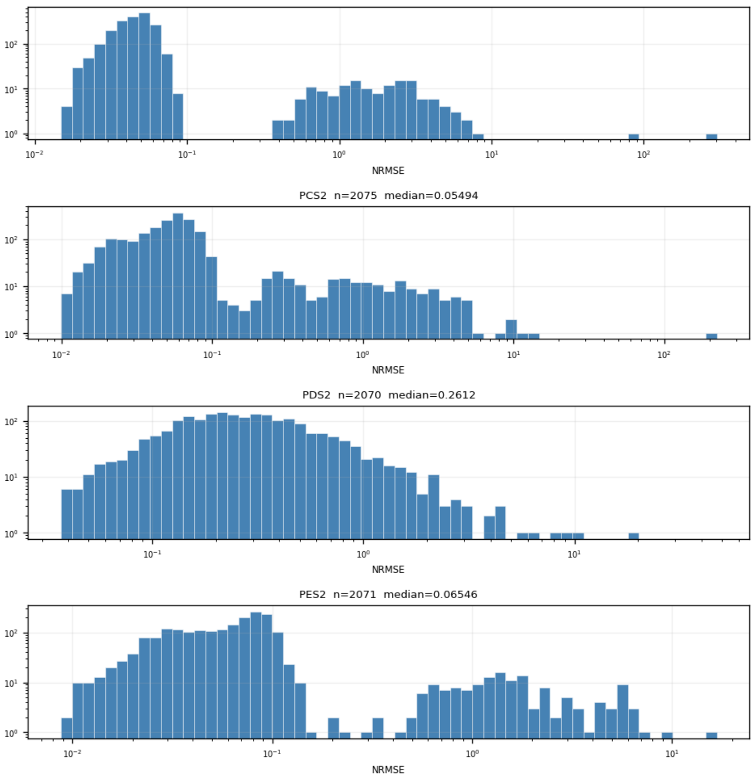
NRMSE = RMS(fit residual)/fitted peak. Bimodal: good-fit population (~0.05–0.1) vs noise triggers (~1–2); the valley at ≈0.4 sets the cut.



part 1/3 (top to bottom)



part 2/3 (top to bottom)

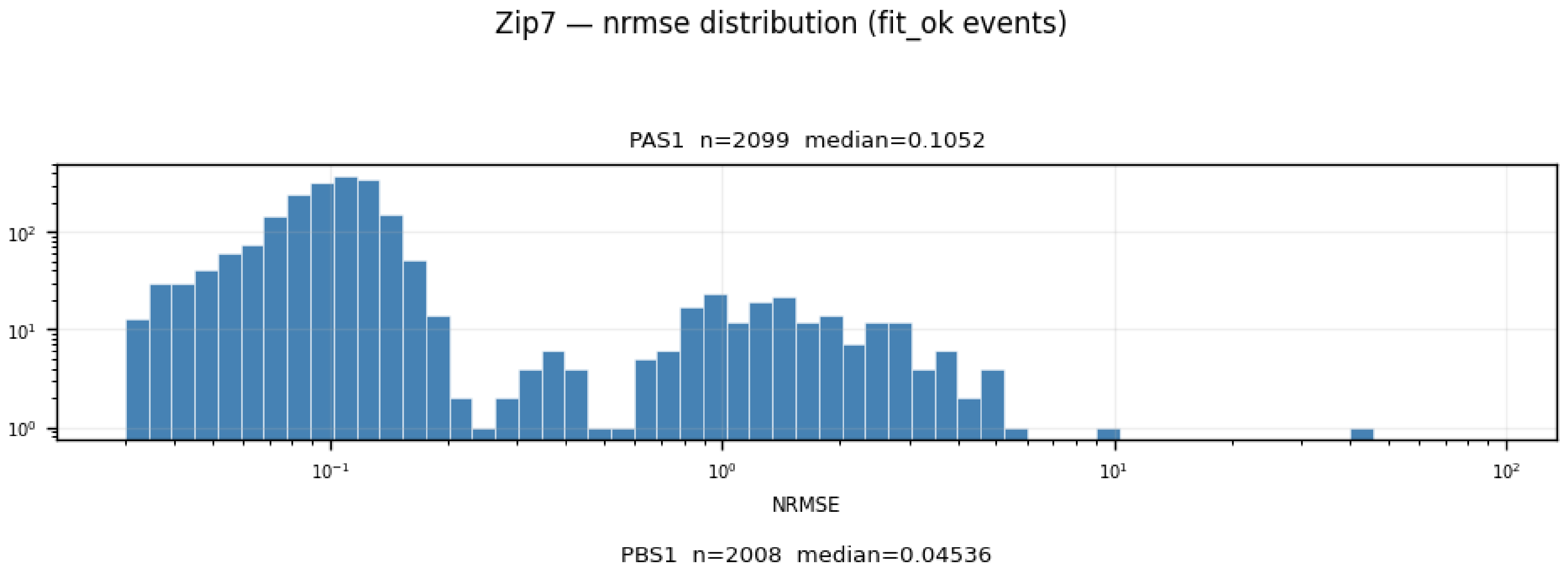


part 3/3 (top to bottom)

Backup — Z7 · nrmse (detail 1:1)

top region of the same figure at full resolution — whole figure on the previous slide

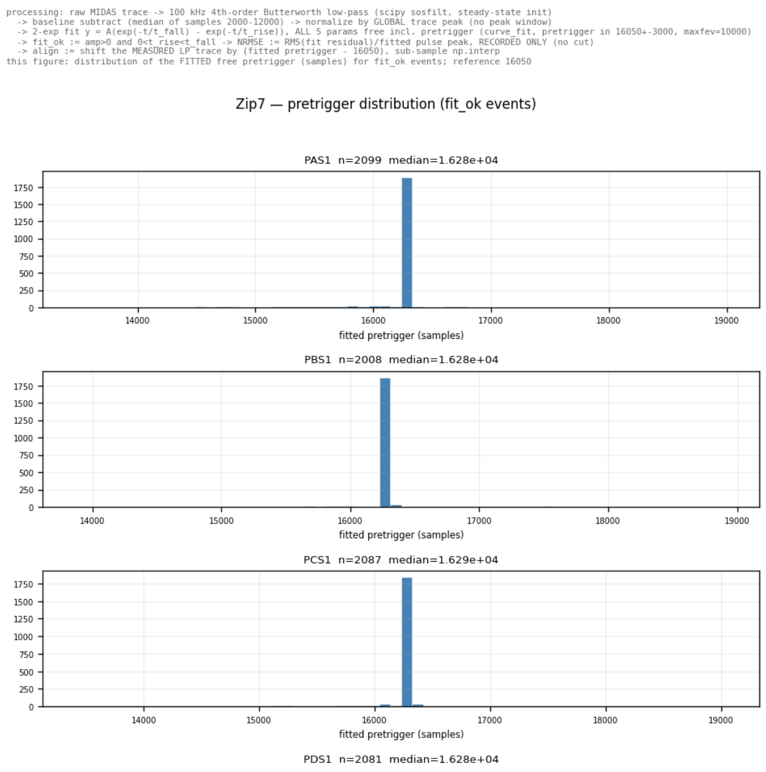
```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
  -> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
  -> 2-exp fit y = A(exp(-t/t_fall) - exp(-t/t_rise)), ALL 5 params free incl. pretrigger (curve_fit, pretrigger in 16050+-3000, maxfev=10000)
  -> fit_ok := amp>0 and 0<t_rise<t_fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
  -> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: NRMSE distribution of fit_ok events, log-log axes (log-spaced bins, log counts); bimodal = good-fit population vs
  noise triggers; valley = natural cut candidate; NO cut applied
```



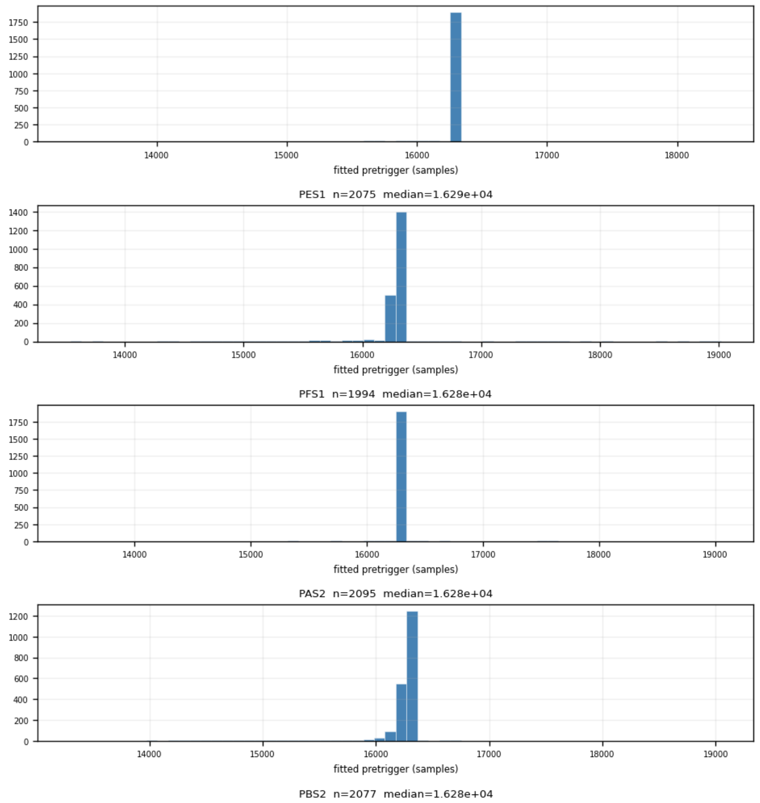
Backup — Z7 · pretrigger

fitted onset distribution · figure: zip7_pretrigger.png

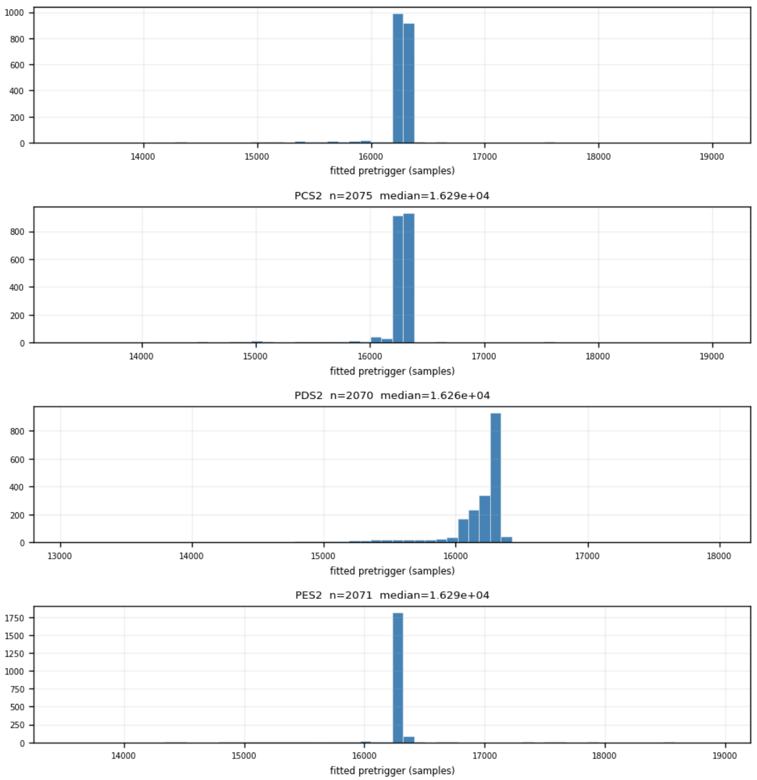
The onset is a FREE fit parameter; fitted values cluster ≈230 samples after the nominal 16050, which is why pinning it inside the fit was wrong.



part 1/3 (top to bottom)



part 2/3 (top to bottom)

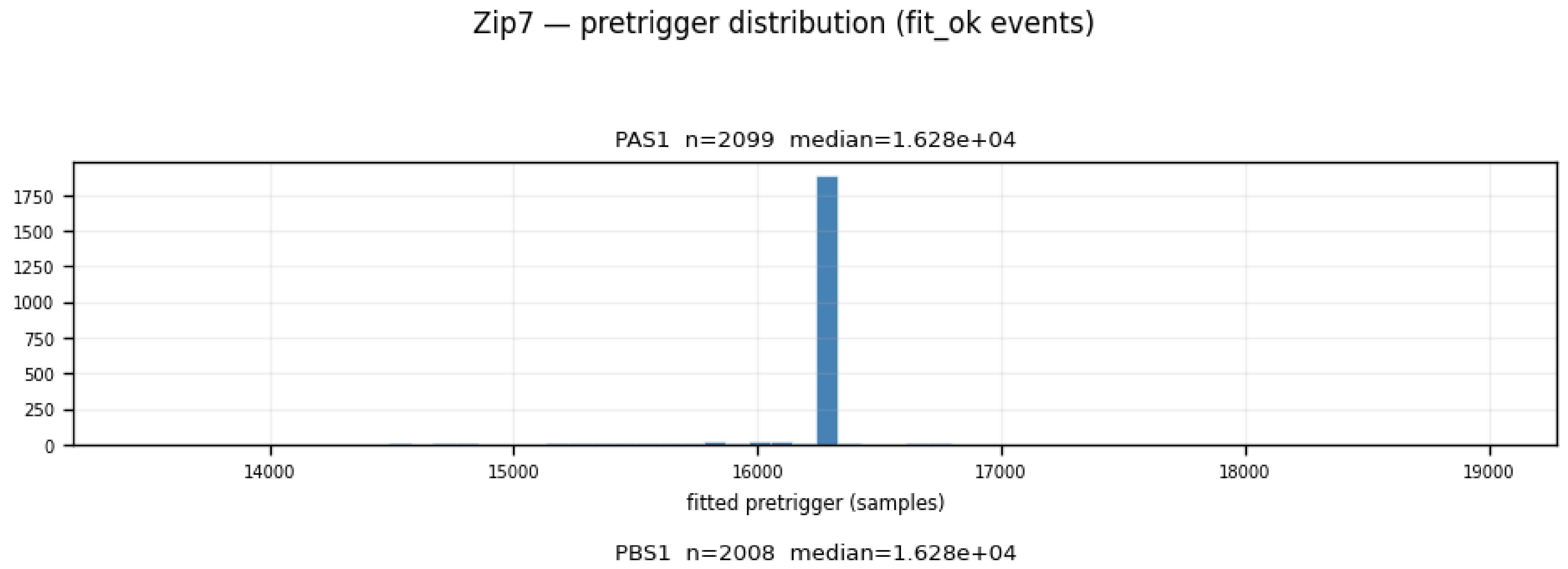


part 3/3 (top to bottom)

Backup — Z7 · pretrigger (detail 1:1)

top region of the same figure at full resolution — whole figure on the previous slide

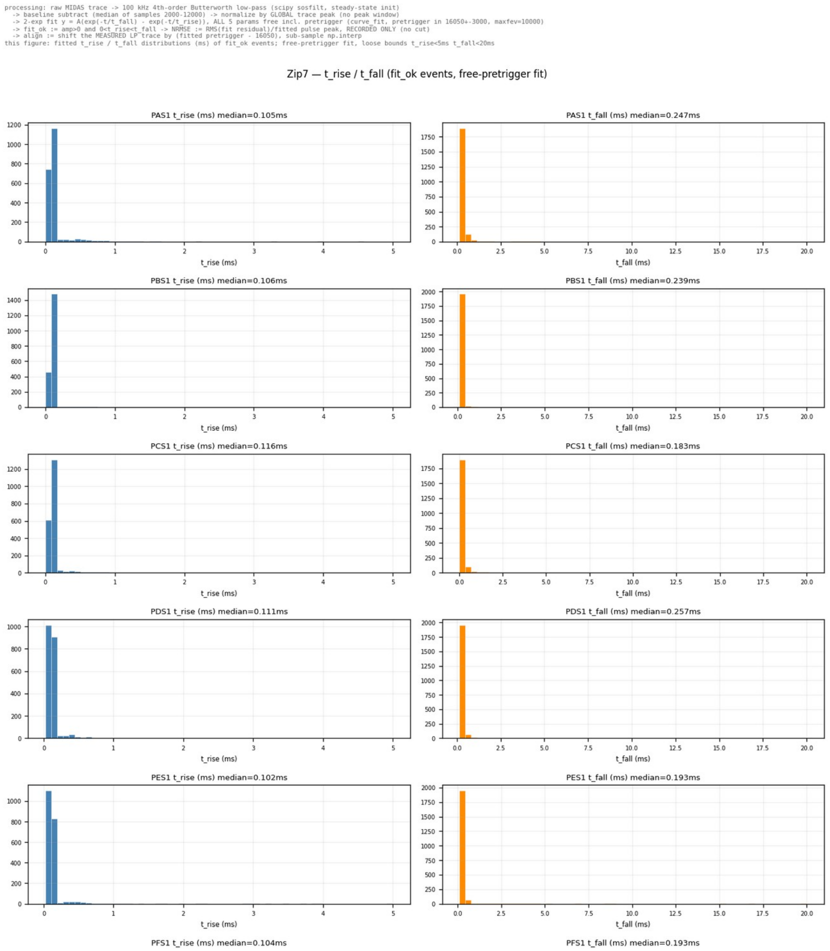
```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
  -> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
  -> 2-exp fit y = A(exp(-t/t_fall) - exp(-t/t_rise)), ALL 5 params free incl. pretrigger (curve_fit, pretrigger in 16050+-3000, maxfev=10000)
  -> fit_ok := amp>0 and 0<t_rise<t_fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
  -> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: distribution of the FITTED free pretrigger (samples) for fit_ok events; reference 16050
```



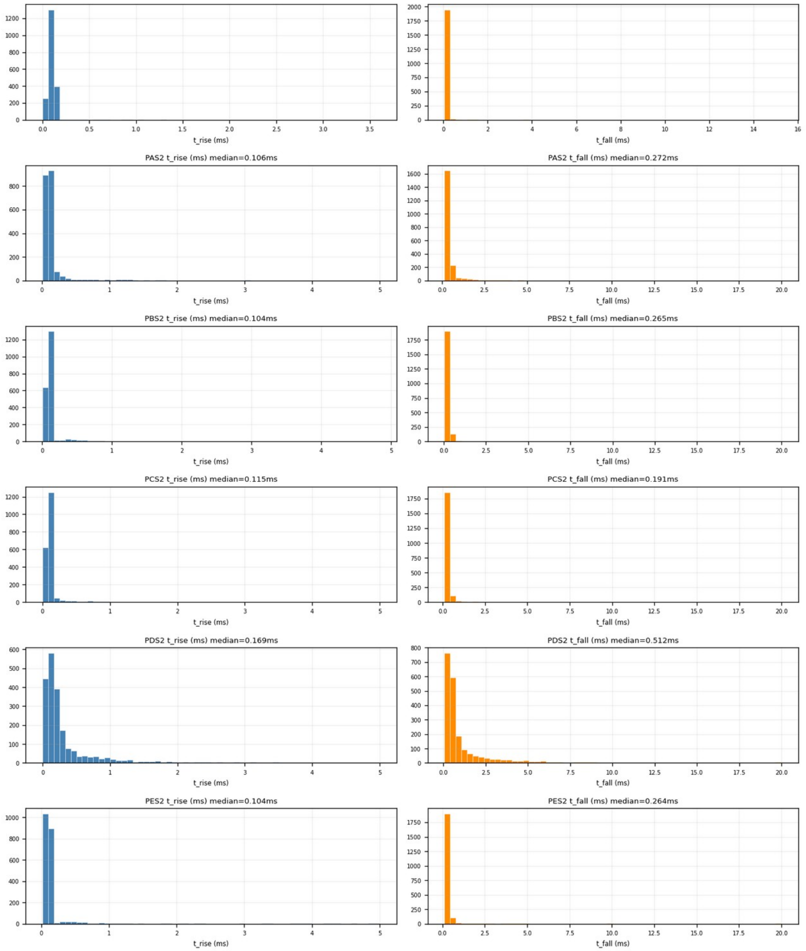
Backup — Z7 · time_constants

fitted τ_rise / τ_fall distributions · figure: zip7_time_constants.png

Fast-pulse population sits at τ_rise ≈ 0.1 ms; isolated spikes at the parameter bounds (5 ms / 20 ms) are fits pinned at the fit limits (noise), removed by the NRMSE cut.



part 1/2 (top to bottom)

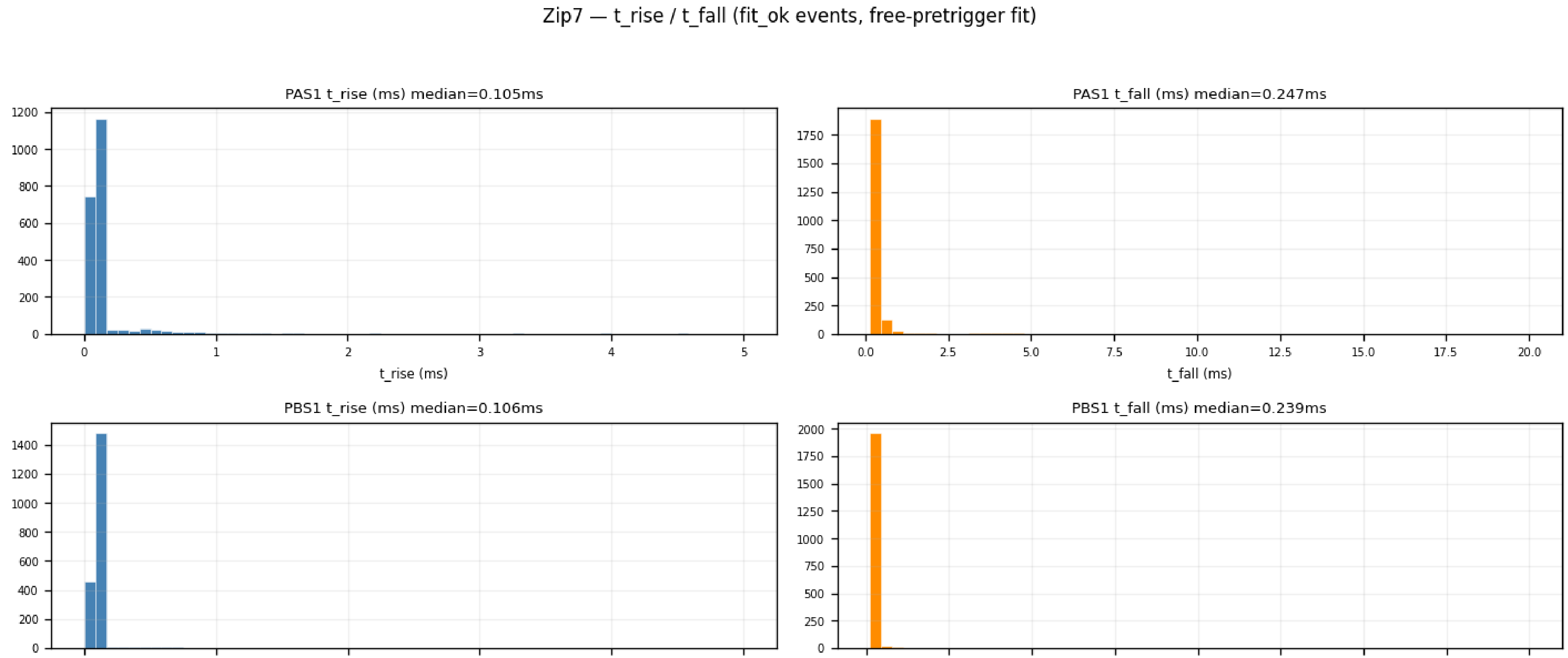


part 2/2 (top to bottom)

Backup — Z7 · time_constants (detail 1:1)

top region of the same figure at full resolution — whole figure on the previous slide

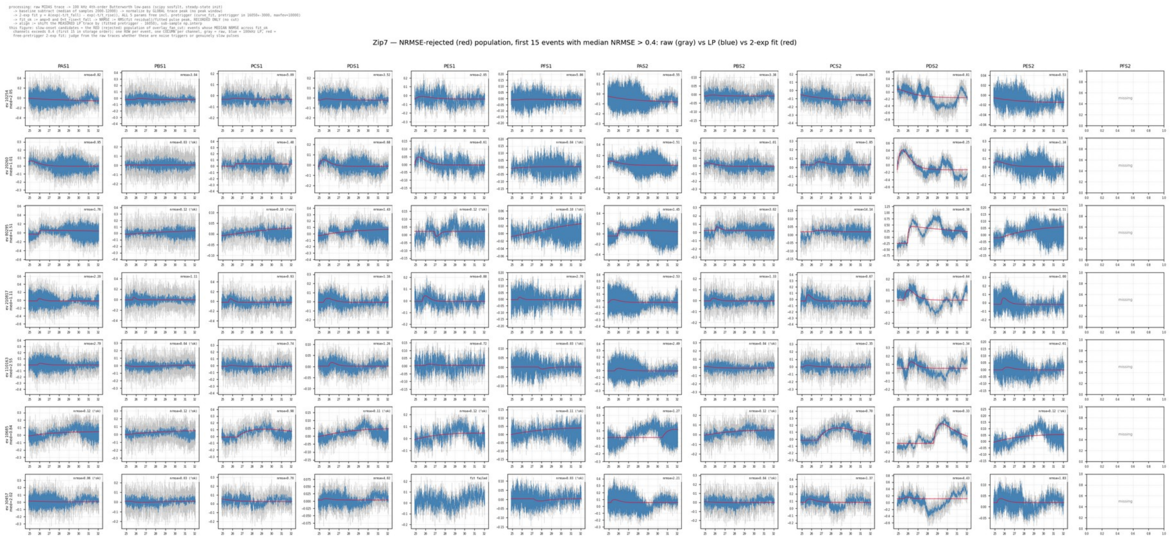
```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
  -> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
  -> 2-exp fit y = A(exp(-t/t_fall) - exp(-t/t_rise)), ALL 5 params free incl. pretrigger (curve_fit, pretrigger in 16050+-3000, maxfev=10000)
  -> fit_ok := amp>0 and 0<t_rise<t_fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
  -> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: fitted t_rise / t_fall distributions (ms) of fit_ok events; free-pretrigger fit, loose bounds t_rise<5ms t_fall<20ms
```



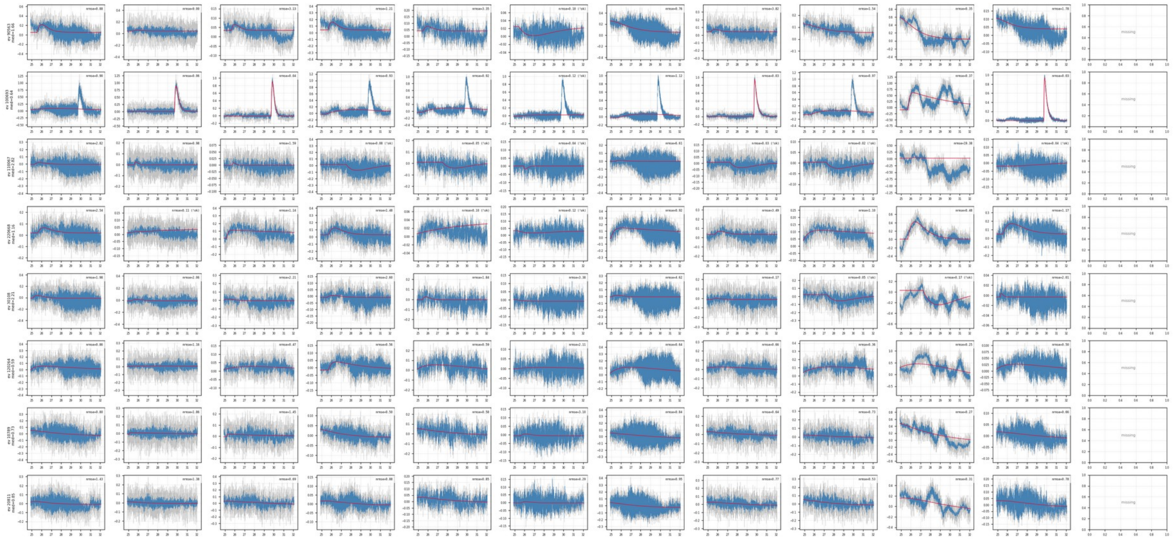
Backup — Z7 · slow_rise_events

the NRMSE-rejected population · figure: zip7_slow_rise_events.png

Events whose median NRMSE across channels > 0.4. Raw traces show no pulse in any channel: the rejected population is noise triggers, not physics.



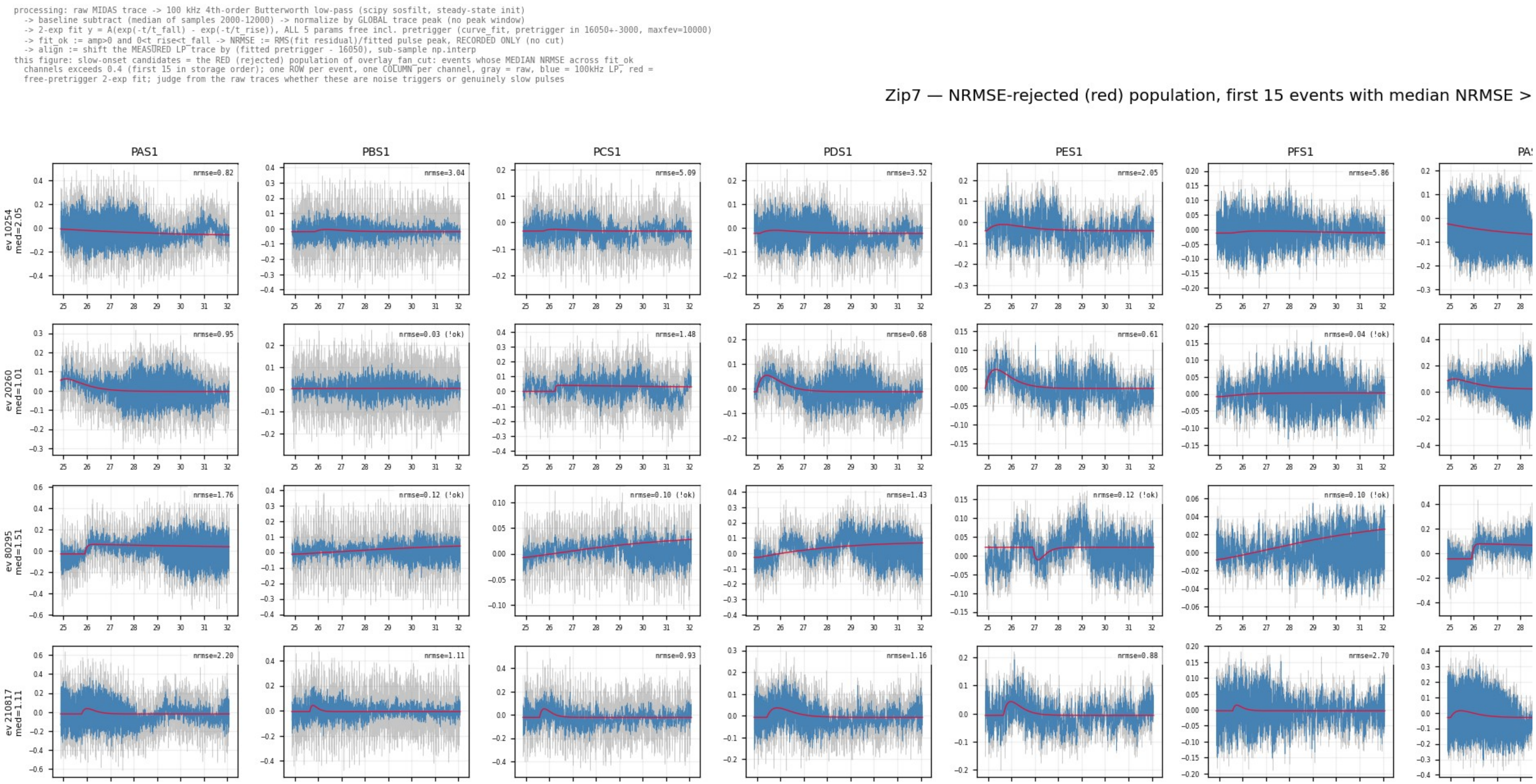
part 1/2 (top to bottom)



part 2/2 (top to bottom)

Backup — Z7 · slow_rise_events (detail 1:1)

top region of the same figure at full resolution — whole figure on the previous slide



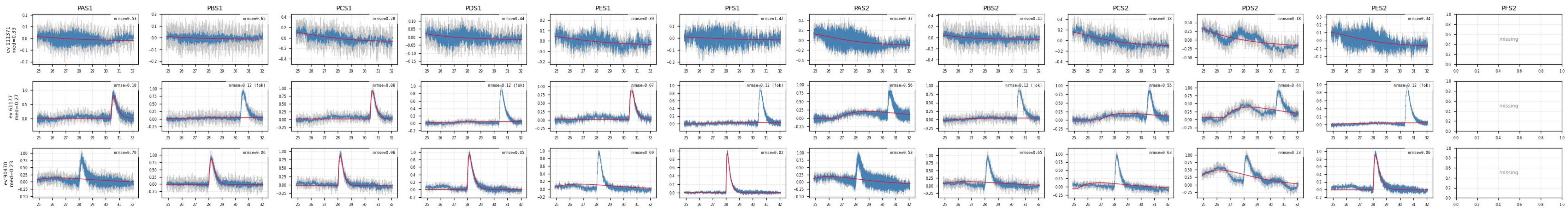
Backup — Z7 · shadow_events

well-fit slow-rise (echo-trigger) events · figure: zip7_shadow_events.png

Median NRMSE ≤ 0.4 AND median τ_rise > 0.2 ms: genuinely slow, well-fit pulses — the faint displaced bundle seen in aligned overlays; kept, and exactly what NxM multi-templates are for.

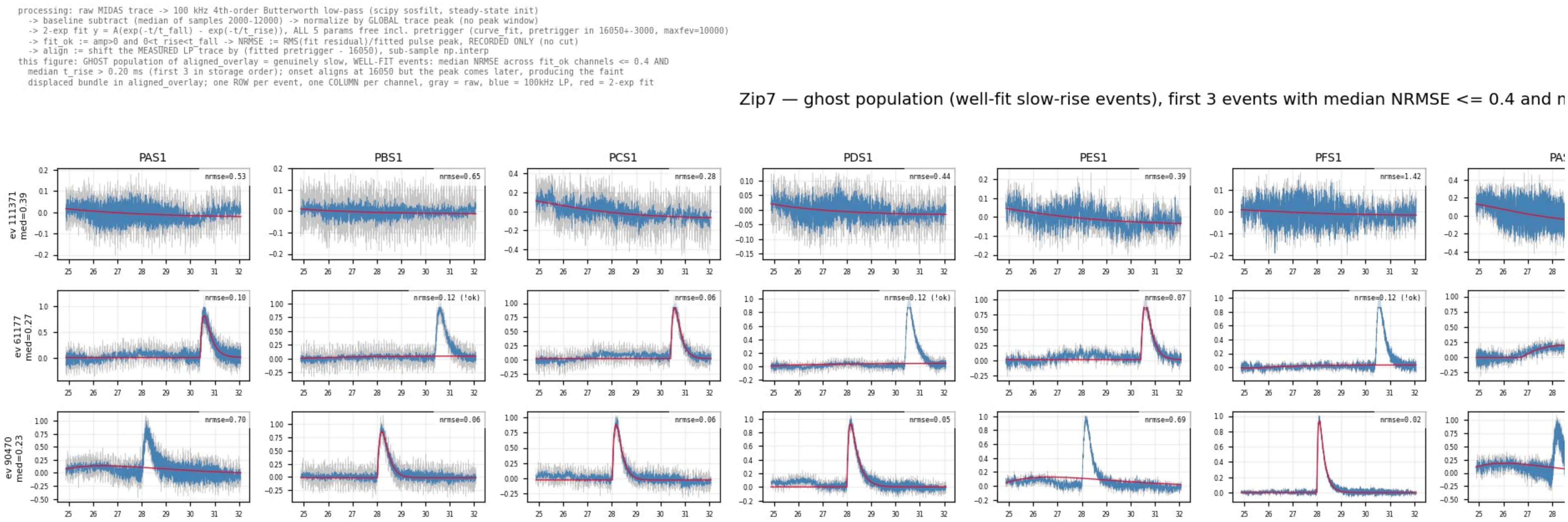
```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
  -> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
  -> 2-exp fit y = A(exp(-t/t_fall) - exp(-t/t_rise)), ALL 5 params free incl. pretrigger (curve_fit, pretrigger in 16050+-3000, maxfev=10000)
  -> fit_ok := amp>0 and 0<t_rise<t_fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
  -> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: GHOST population of aligned_overlay = genuinely slow, WELL-FIT events: median NRMSE across fit_ok channels <= 0.4 AND
  median t_rise > 0.20 ms (first 3 in storage order); onset aligns at 16050 but the peak comes later, producing the faint
  displaced bundle in aligned_overlay; one ROW per event, one COLUMN per channel, gray = raw, blue = 100kHz LP, red = 2-exp fit
```

Zip7 — ghost population (well-fit slow-rise events), first 3 events with median NRMSE <= 0.4 and median t_rise > 0.20 ms: raw (gray) vs LP (blue) vs 2-exp fit (red)



Backup — Z7 · shadow_events (detail 1:1)

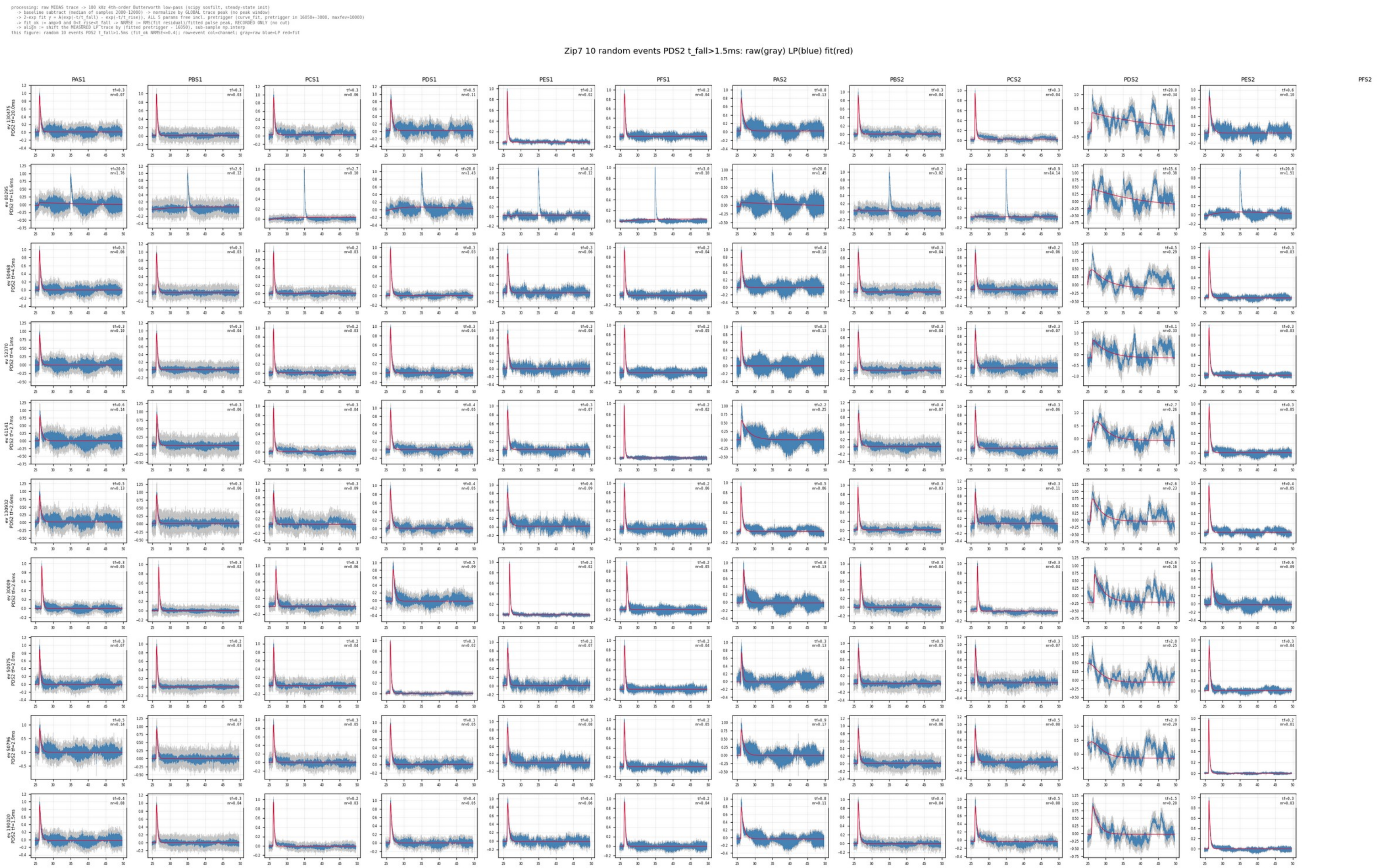
top region of the same figure at full resolution — whole figure on the previous slide



Backup — Z7 · slow_fall_events

long-τ_fall study (PDS2 reference) · figure: zip7_PDS2_slow_fall.png

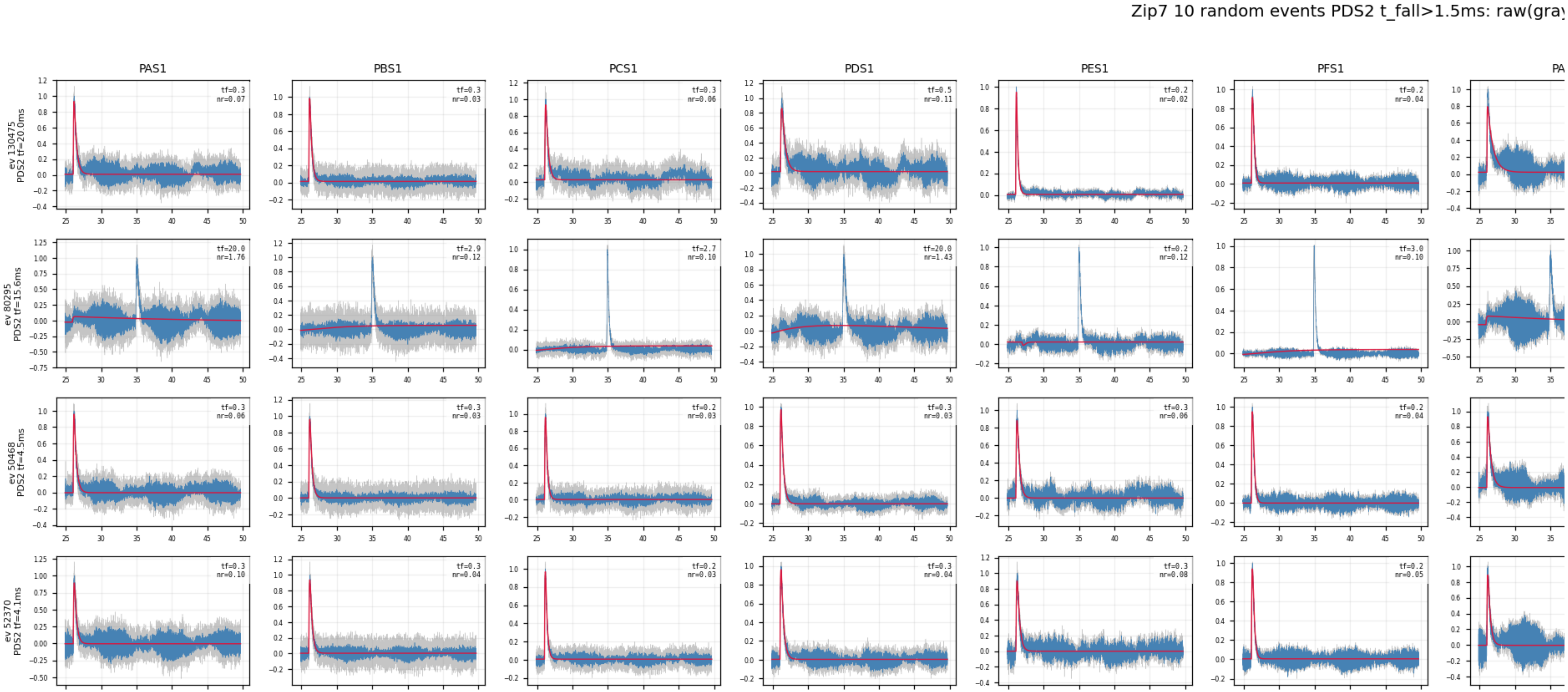
10 random events with PDS2 τ_fall > 1.5 ms drawn in every channel: 11 channels show normal fast pulses; the long fall is a PDS2-only low-frequency artifact — not slow physics, no event-level cut.



Backup — Z7 · slow_fall_events (detail 1:1)

top region of the same figure at full resolution — whole figure on the previous slide

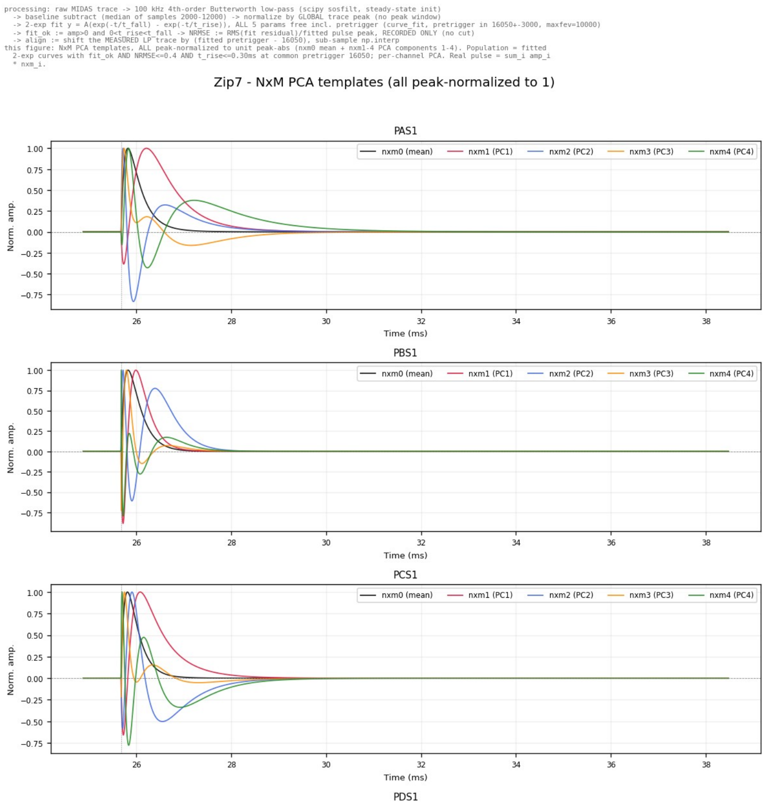
```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
  -> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
  -> 2-exp fit y = A(exp(-t/t_fall) - exp(-t/t_rise)), ALL 5 params free incl. pretrigger (curve_fit, pretrigger in 16050+-3000, maxfev=10000)
  -> fit_ok := amp>0 and 0<t_rise<t_fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
  -> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: random 10 events PDS2 t_fall>1.5ms (fit_ok NRMSE<=0.4); row=event col=channel; gray=raw blue=LP red=fit
```



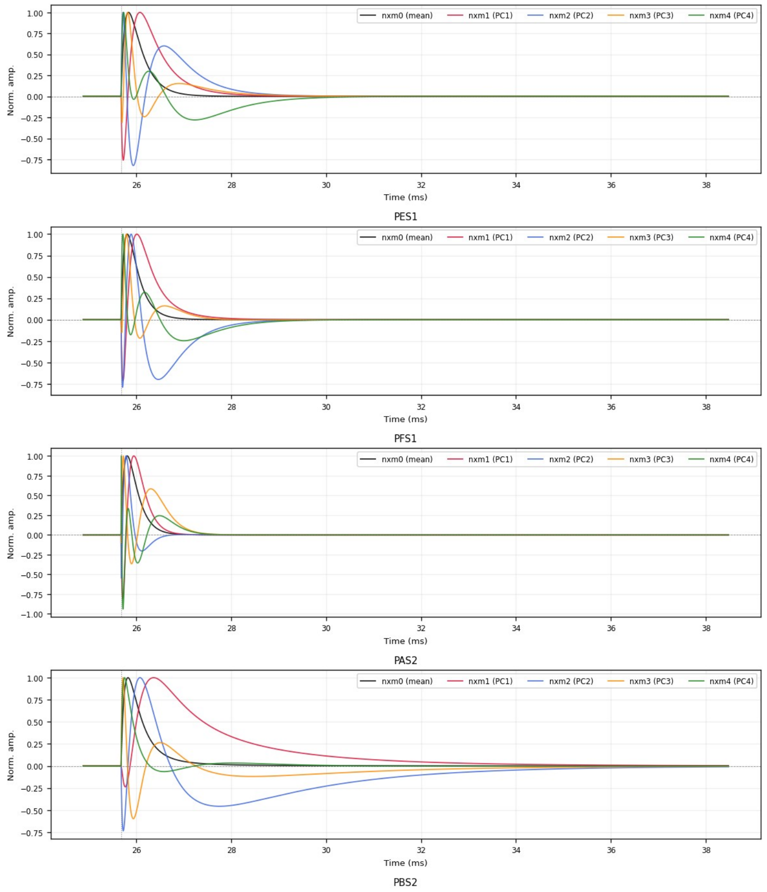
Backup — Z7 · deliverables/nxm

PCA templates nxm0–4 · figure: zip7_pca_templates.png

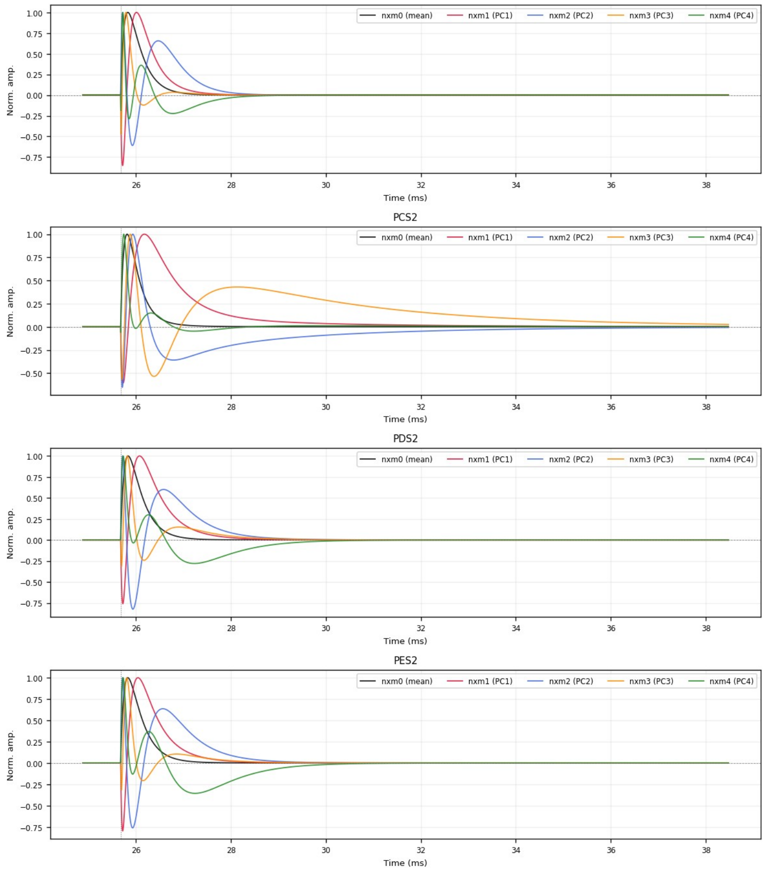
nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized to 1; PC1+PC2 capture 96–98% of the shape variance.



part 1/3 (top to bottom)



part 2/3 (top to bottom)

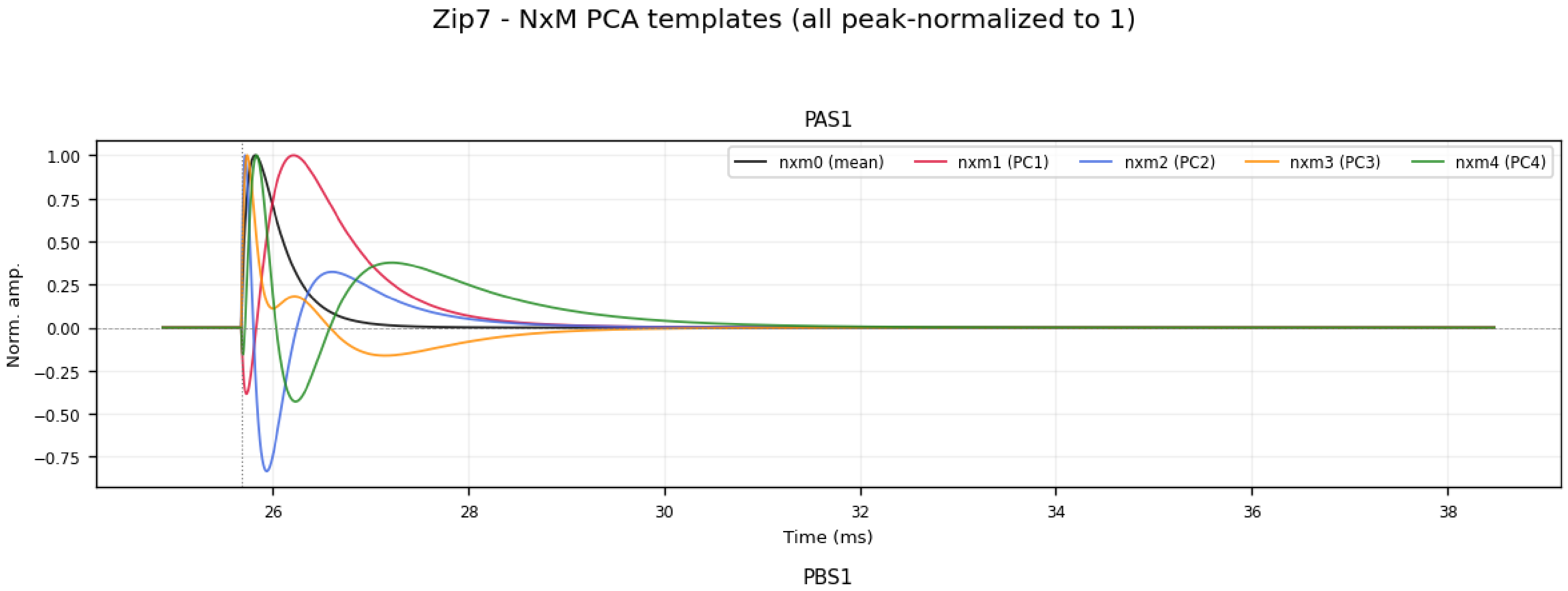


part 3/3 (top to bottom)

Backup — Z7 · deliverables/nxm (detail 1:1)

top region of the same figure at full resolution — whole figure on the previous slide

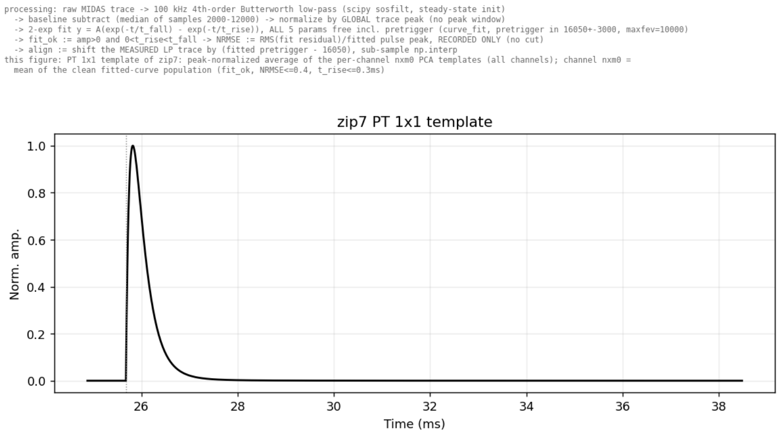
```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
  -> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
  -> 2-exp fit y = A(exp(-t/t_fall) - exp(-t/t_rise)), ALL 5 params free incl. pretrigger (curve_fit, pretrigger in 16050+-3000, maxfev=10000)
  -> fit_ok := amp>0 and 0<t_rise<t_fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
  -> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: NxM PCA templates, ALL peak-normalized to unit peak-abs (nxm0 mean + nxm1-4 PCA components 1-4). Population = fitted
  2-exp curves with fit_ok AND NRMSE<=0.4 AND t_rise<=0.30ms at common pretrigger 16050; per-channel PCA. Real pulse = sum_i amp_i
  * nxm_i.
```



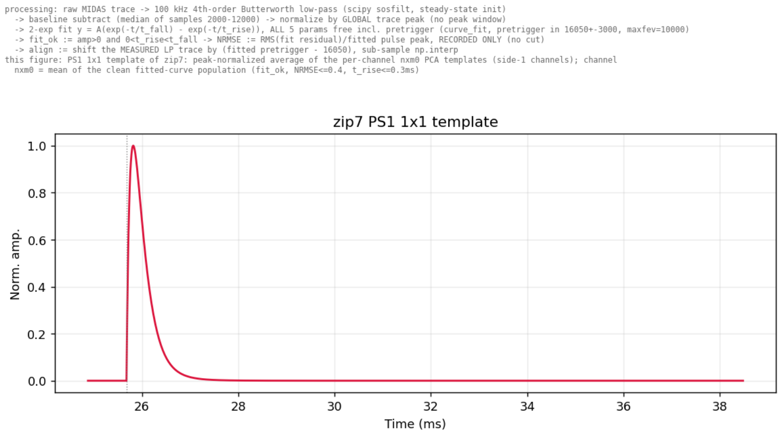
Backup — Z7 · 1x1 summed templates PT / PS1 / PS2

figures: deliverables/1x1/plots/{PT,PS1,PS2}/zip7_*.png

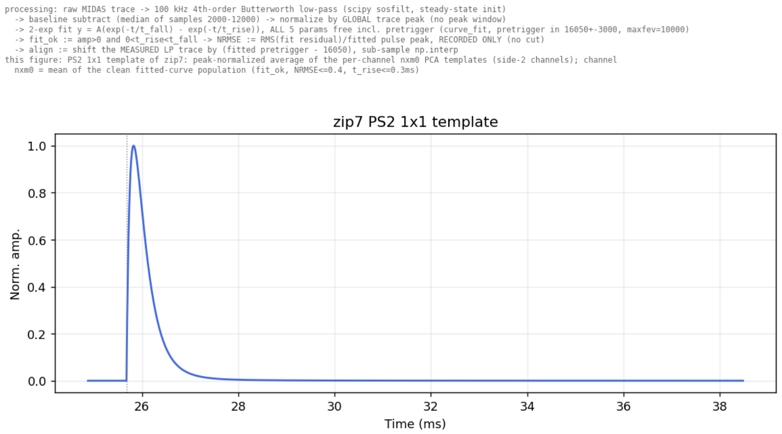
Single-template (1x1) summed curves: peak-normalized average of the per-channel nxm0 templates — PT = all channels, PS1 / PS2 = side-1 / side-2 only.



Z7 PT



Z7 PS1



Z7 PS2